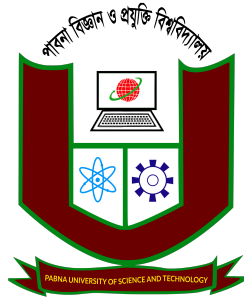


DG-SSOFNet: A Novel Corpus and Cross-Attention Multi-Stream Fusion with Contrastive Learning for Bangla Fake News Detection



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Prepared By:

Shubail Haque Turza

Roll No: 210606

Reg. No: 1065411

Session: 2020–2021

Supervised By:

Dr. Md. Sarwar Hosain

Professor

**Department of Information and Communication Engineering
Pabna University of Science and Technology
Pabna-6600, Bangladesh
August 2026**

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Dedicated to...

My family

and

society.

Declaration

I hereby declare that the thesis entitled “**DG-SSOFNet: A Novel Corpus and Cross-Attention Multi-Stream Fusion with Contrastive Learning for Bangla Fake News Detection**” was completed by me under the course entitled “**Thesis**” (Course Code: **ICE - 4210**). I confirm that this work is my own academic work and has not been submitted, either in full or in part, to any other institution for any degree or diploma, except where proper acknowledgement has been made.

Shubail Haque Turza

Roll No.: 210606

Reg. No.: 1065411

Session: 2020–2021

Department of Information and Communication Engineering

Pabna University of Science and Technology

Pabna-6600, Bangladesh

Certificate of Approval

This is to certify that the thesis entitled “**DG-SSOFNet: A Novel Corpus and Cross-Attention Multi-Stream Fusion with Contrastive Learning for Bangla Fake News Detection**” submitted by **Shubail Haque Turza** (Roll No: 210606, Reg. No: 1065411, Session: 2020–2021) in partial fulfillment of the requirements for the degree of Bachelor of Science in Engineering in Information and Communication Engineering at Pabna University of Science and Technology, has been carried out under my supervision. The presentation of this thesis is satisfactory in terms of language, format, and content.

Dr. Md. Sarwar Hosain

Professor

Department of Information and Communication Engineering

Pabna University of Science and Technology

Pabna-6600, Bangladesh

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Shubail Haque Turza

August 2026

Pabna, Bangladesh

Abstract

Detecting fake news in Bangla is challenging for two main reasons. First, clean datasets are limited. Second, titles collected from fact-checking websites often contain editorial phrases such as “fact check”, “false claim”, or “rumor alert”. A standard classifier can exploit these phrases to achieve high accuracy without learning the meaning of the news claim. To address this problem, this thesis builds a cleaned Bangla fake news corpus and evaluates a multi-stream classification model named DG-SSOFNet (Decontaminated Generative - Semantic, Syntactic, and Orthographic Fusion Network). The corpus contains 43,860 unique titles, comprising 33,147 real-news titles and 10,713 reconstructed fake claims. We designed a multi-pass reconstruction pipeline to remove 27 categories of verdict-leaking terms before model training. After applying a stratified group split and removing 132 near-duplicate evaluation items, the final benchmark contains 43,728 titles. On the modeling side, DG-SSOFNet combines three views of a news title. A pre-trained BanglaBERT encoder captures meaning, a Part-of-Speech (POS) guided network models grammatical sentence structure, and a character-level CNN captures spelling patterns. These feature streams are projected to a unified dimension and integrated through a cross-attention gating block. The network is trained using focal loss and supervised contrastive learning (SupCon) to improve class separation. Experiments show that DG-SSOFNet without data augmentation achieves a mean test Macro-F1 of 0.9711, closely followed by standalone BanglaBERT at 0.9696. With data augmentation, DG-SSOFNet reaches 0.9695 compared to 0.9685 for BanglaBERT. Statistical significance testing using paired confidence intervals and exact McNemar tests ($p = 0.558$) demonstrates that DG-SSOFNet performs similarly to Transformer baselines. However, it offers practical advantages in probability calibration and robustness against typographical errors. The primary contributions of this thesis are the decontaminated Bangla corpus, the reproducible evaluation framework, and an assessment of where multi-stream fusion helps in low-resource fake news detection.

Keywords: Fake News Detection, Bangla NLP, Transformer, Contrastive Learning, Multi-Stream Fusion, Fact-Checking.

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List of Abbreviations

Abbreviation	Full Form / Meaning
AI	Artificial Intelligence
BERT	Bidirectional Encoder Representations from Transformers
BiGRU	Bidirectional Gated Recurrent Unit
BTRC	Bangladesh Telecommunication Regulatory Commission
CharCNN	Character-Level Convolutional Neural Network
CNN	Convolutional Neural Network
DG-SSOFNet	Decontaminated Generative - Semantic, Syntactic, and Orthographic Fusion Network
ECE	Expected Calibration Error
FNN	Feedforward Neural Network
HTML	HyperText Markup Language
LLM	Large Language Model
LSTM	Long Short-Term Memory
MCC	Matthews Correlation Coefficient
MLP	Multi-Layer Perceptron
NLP	Natural Language Processing
OOD	Out-of-Distribution
POS	Part-of-Speech
PR-AUC	Precision-Recall Area Under Curve
RNN	Recurrent Neural Network
ROC	Receiver Operating Characteristic
ROC-AUC	Receiver Operating Characteristic Area Under Curve
SupCon	Supervised Contrastive Learning
SVM	Support Vector Machine
TextCNN	Text Convolutional Neural Network
TF-IDF	Term Frequency-Inverse Document Frequency
URL	Uniform Resource Locator

List of Publications

Conference / Journal Papers:

1. **S. H. Turza**, A. Mahdi, M. I. S. Shad, M. H. Kabir, and A. Islam, “Fake News Detection in Bangla: A Transformer Ensemble Framework for Robust Classification,” in *2025 28th International Conference on Computer and Information Technology (ICCIT)*, Cox’s Bazar, Bangladesh, 2025, doi: <https://doi.org/10.1109/ICCIT68739.2025.11491228>.
2. N. T. Susmi, M. C. Chanda, M. S. Hosain, M. R. Hossen, M. A. Hossain, A. F. M. Z. Abadin, and **S. H. Turza**, “Enhancing DeepFake Classification Performance Using a CNN and XceptionNet-Based Pipeline,” in *2025 IEEE 2nd International Conference on Computing, Applications and Systems (COMPAS)*, 2025, doi: <https://doi.org/10.1109/COMPAS67506.2025.11381636>.
3. **S. H. Turza** and M. S. Hosain, “DG-SSOFNet: A Novel Corpus and Cross-Attention Multi-Stream Fusion with Contrastive Learning for Bangla Fake News Detection,” 2026. (*Manuscript under preparation*).

Introduction

1.1 Motivation

The rapid expansion of the internet and the use of smartphones have changed how people get information. Social media is now the main source of news for many people. While this allows for instant access to events and easier information sharing, it also creates an environment where fake news spreads quickly. Fake news is not new, but digital platforms make it spread faster and reach more people than before [1].

This is a serious problem in Bangladesh. With over 180 million people, digital connectivity has grown rapidly in the country. There are over 50 million active Facebook users in Bangladesh [2], and for many, social media is their only news source. Since these platforms lack traditional editorial review, false or biased content often goes viral. This spread is driven by algorithms that reward user engagement and shocking content over factual accuracy.

The spread of fake news online in Bangladesh causes real harm. Misinformation has led to communal violence, causing deaths and property damage [3]. During elections, disinformation campaigns try to manipulate voters. During the COVID-19 pandemic, fake cures and anti-vaccine content harmed public health and interfered with government responses [4]. These examples show why we need better technology to detect and stop fake news.

Researchers are using Natural Language Processing (NLP) and machine learning to find fake news automatically. There is a lot of progress in high-resource languages like English, which have large datasets and pre-trained models [5]. However, Bangla is still a low-resource language for NLP. Even though many people speak it, there are not enough large, high-quality datasets for fake news detection [6].

A major problem with current Bangla fake news datasets is their quality. Many are built by scraping fact-checking websites, which often use titles with obvious labels like "Fact Check: [Claim] is False" or "Rumor Alert: [Claim]". When models train on these scraped titles, they just learn to look for those label words instead of understanding the actual text. This issue, called verdict wrapper leakage, makes the models look highly accurate in testing, but they fail in real-world situations where the labels are missing.

This project aims to fill that gap. We need fake news detection systems for Bangla that work well on new data. First, we must fix the dataset issue by creating a corpus that looks like real fake news without artificial labels. Second, because deceptive language is complicated, basic semantic models are not enough. We need architectures that combine different types of linguistic information to make better predictions.

1.2 Problem Statement

The main problem this thesis addresses is that current Bangla fake news detection models fail on real-world data because of biased datasets and the limitations of single-stream architectures.

The first issue is how the datasets are built. Fact-checking groups format their headlines to show if a claim is true or false right away. Datasets made from these sources are full of debunking terms. For example, a title might say "Fact Check: The video showing a bridge collapse is from 2015, not recent." If a dataset has many fake examples like this alongside true examples from news sites without these terms, a machine learning model will learn to classify any text containing "Fact Check" or "is from" as fake.

This means the model acts as a keyword spotter rather than analyzing the meaning of the text. It learns shortcuts instead of actual deceptive language patterns. These models might get over 90% accuracy on their test sets, but they fail when they see a new, deceptive claim on social media without a fact-checking wrapper. No study has fixed this leakage issue for Bangla NLP, and there is no large effort to clean existing datasets so models learn real signs of deception.

Second, the models currently used for Bangla fake news detection are often too simple. Most rely on single-stream models, passing text through one language model. Semantic models like BanglaBERT capture meaning well, but they can miss other linguistic signals. Fake news often has strange grammar or unusual character patterns that single-stream models miss. Using only one type of feature representation means these systems miss information needed for better detection.

Therefore, the problem has two parts. First, the data used to train the models is

flawed due to verdict-wrapper leakage. Second, the single-stream models commonly used cannot capture the complex nature of deceptive language in Bangla. We need a clean dataset and a multi-stream architecture that can learn complex patterns to build an effective fake news detection system.

1.3 Research Questions

To build a better fake news detection system for Bangla, this thesis asks the following research questions:

RQ1: Can we build a Bangla fake news corpus that forces models to learn actual linguistic features of fake news by removing verdict-leaking terms?

Current datasets let models cheat by including obvious clues in the text. This question explores whether using language models to rewrite the data can remove these shortcuts and create a better dataset for testing model generalization in Bangla.

RQ2: Does a multi-stream architecture that combines semantic, syntactic, and character-level features improve detection compared to single-stream models?

This question investigates whether combining semantic meaning (using BanglaBERT) with grammar (Part-of-Speech tags) and character patterns (using Convolutional Neural Networks) provides a better representation of the text and improves accuracy.

RQ3: How does using supervised contrastive learning and cross-attention affect the model’s calibration and its ability to handle noisy data?

This question looks at whether contrastive learning helps the model separate real and fake news better, improving its confidence scores. It also asks whether a cross-attention mechanism helps the model weigh different features, making its decisions clearer and more reliable.

1.4 Research Objectives

Based on the research questions, this thesis has the following objectives:

1. Construct a clean Bangla fake news corpus: The main goal is to create a dataset of 43,860 titles from fact-checking sites. We will use a pipeline to remove verdict-leaking terms. This involves using Large Language Models to rewrite the titles into natural sentences that keep the original meaning but remove the debunking labels, providing clean data for training.

2. Design a cross-attention multi-stream architecture (DG-SSOFNet): We aim to develop a neural network that processes text through multiple streams. This will include a

semantic stream using BanglaBERT, a syntactic stream using Part-of-Speech (POS) tags, and a character-level stream using CharCNN. We will design a cross-attention mechanism to fuse these representations.

3. Add supervised contrastive learning to the training process: We will use a Supervised Contrastive Learning (SupCon) loss function during training. The goal is to pull samples of the same class closer together and push apart samples from different classes (real vs. fake). We aim to show that this improves generalization on unseen data compared to standard cross-entropy loss.

4. Evaluate the model comprehensively: We will evaluate DG-SSOFNet against baseline models using metrics like Accuracy and F1-score, as well as statistical tests (McNemar’s test, Bootstrap Confidence Intervals). We will also check model calibration (using Expected Calibration Error), evaluate its robustness to noisy data, and use LIME to understand how the model makes decisions.

1.5 Contributions

This research makes several contributions to Natural Language Processing for Bangla:

1. A Clean Corpus for Bangla Fake News: We present a dataset of 43,860 clean titles. Unlike previous datasets [6, 7] that have verdict wrapper leakage, this corpus forces models to learn actual language features instead of keywords. We did this by using Large Language Models to rewrite fact-checking headlines into natural claims. We are releasing this dataset publicly to provide a better benchmark for evaluating Bangla fake news models.

2. Cross-Attention Multi-Stream Fusion Architecture (DG-SSOFNet): We propose a deep learning architecture named DG-SSOFNet. It extracts features across three dimensions simultaneously. It uses BanglaBERT [8] for semantics, Part-of-Speech tags for syntax, and a Character Convolutional Neural Network (CharCNN) for orthographic patterns. A cross-attention block helps the model learn the best weights among these streams.

3. Supervised Contrastive Learning Implementation: We add Supervised Contrastive Learning (SupCon) [9] to the training process. SupCon helps create tighter clusters for each class in the latent space, which improves the model’s ability to tell real and fake news apart.

4. Comprehensive Evaluation: We provide a thorough evaluation methodology. We use bootstrap confidence intervals and McNemar’s test to check statistical significance. We analyze model calibration using Expected Calibration Error (ECE) and test ro-

bustness against character-level noise. We also apply Local Interpretable Model-agnostic Explanations (LIME) [10] to show how different streams contribute to the final decisions.

1.6 Organization of the Thesis

This thesis has six chapters:

Chapter 1: Introduction presents the motivation for the study, discusses fake news in Bangladesh, defines the problem of verdict wrapper leakage, lists the research questions and objectives, and summarizes the contributions.

Chapter 2: Literature Review covers existing research in fake news detection, including traditional machine learning, deep learning, and transformer models. It reviews studies on the Bangla language and discusses their datasets and limitations.

Chapter 3: Dataset Construction and Analysis details how we created the clean corpus. It explains the problem of verdict wrappers and describes the LLM reconstruction pipeline. It also includes exploratory data analysis of the new dataset.

Chapter 4: Methodology introduces the DG-SSOFNet architecture. It details the three parallel streams (BanglaBERT, POS tagging, CharCNN), explains the cross-attention mechanism, and describes the supervised contrastive learning objective.

Chapter 5: Experiments and Results presents the evaluation of the model. It covers the experimental setup, baseline models, metrics, and quantitative results. It also includes analyses of calibration, robustness, and interpretability using LIME.

Chapter 6: Conclusion and Future Work summarizes the findings, discusses their implications, and suggests directions for future research in Bangla fake news detection.

Background Study and Literature Review

2.1 Fake News: Definition, Types, and Impact

The rapid growth of social media has fundamentally changed how people consume and share information. While these platforms allow for fast communication, they also make it easy to spread false information. Researchers define fake news as news articles that are intentionally and verifiably false and could mislead readers [1]. However, the term covers a wide range of deceptive content.

Information disorder usually falls into three main categories: misinformation, disinformation, and malinformation [11]. Misinformation refers to false information shared without harmful intent. Disinformation is false information created and shared specifically to cause harm or deceive people. Malinformation involves genuine information shared to cause harm, such as leaks or hate speech. Fake news generally aligns with disinformation.

Fake news takes many forms. Researchers often group these forms based on how they mislead [12].

- **Satire or Parody:** Content created for humor but sometimes mistaken for real news.
- **Misleading Content:** True information framed in a way that misleads the reader, often by using sensational headlines that do not match the article body.
- **Fabricated Content:** Completely made-up news stories designed to deceive.

The impact of fake news is global. It influences elections, harms public health, and damages personal reputations. During political events, fake news can polarize voters and

damage democratic processes [13]. During the COVID-19 pandemic, false health claims led to dangerous behaviors and vaccine hesitancy [14].

In Bangladesh, the impact of fake news is particularly severe due to rapidly growing internet access combined with low digital literacy. Social media platforms like Facebook serve as primary news sources for many citizens. This environment allows rumors to spread out of control. For example, in 2019, rumors spread on Facebook claiming that human heads were needed to build the Padma Bridge. This false claim led to mob violence and the deaths of several innocent people [15]. Similarly, politically motivated fake news frequently surfaces during national elections to manipulate public opinion. These real-world consequences highlight the urgent need for automated systems that can accurately detect fake news in Bangla.

2.2 Natural Language Processing for Low-Resource Languages

Natural Language Processing (NLP) Models require large amounts of data to learn effectively. High-resource languages like English have massive text corpora, extensive annotated datasets, and well-developed linguistic tools. In contrast, low-resource languages lack these essential elements, making NLP tasks much more difficult [16].

A language is considered low-resource if it lacks large unannotated text corpora, has limited task-specific labeled data, or lacks basic processing tools such as tokenizers and stemmers. Many languages spoken by millions of people fall into this category.

Bangla is the seventh most spoken language in the world, with over 260 million speakers. Despite this, it remains a low-resource language in NLP. Several linguistic and technical challenges make processing Bangla text difficult.

First, Bangla has a highly complex morphology. It is an agglutinative language, meaning words are formed by combining multiple morphemes. A single root word can have hundreds of variations depending on tense, person, and case. This high vocabulary explosion makes standard word-level tokenization inefficient.

Second, the language features complex spelling variations and a high number of compound characters (juktakkhor). People also frequently use informal spelling and mix Bangla with English words (code-mixing) on social media, adding to the noise [17].

Third, there is a severe lack of high-quality, large-scale datasets for specific tasks like sentiment analysis, named entity recognition, and fake news detection.

Despite these challenges, the Bangla NLP community has made progress. Early ef-

2.3 Text Classification Approaches

forts focused on creating basic resources, such as stop-word lists and rule-based stemmers. Later, researchers trained distributed word embeddings such as Word2Vec and fastText specifically on the Bangla Wikipedia and news corpora [18].

A major improvement came with the introduction of pre-trained language Models. While multilingual Models such as mBERT [19] and XLM-R [20] include Bangla in their training data, they often underperform compared to language-specific Models. To address this, researchers developed BanglaBERT [8], a BERT-based Model pre-trained exclusively on a large, clean Bangla corpus. Models like BanglaBERT have set new baselines for almost all downstream Bangla NLP tasks, moving the field away from traditional machine learning towards advanced deep learning.

2.3 Text Classification Approaches

Fake news detection is primarily framed as a text classification task, where a Model takes a news article as input and predicts whether it is real or fake. Researchers have applied various methods to this problem, ranging from simple rule-based systems to complex neural networks.

2.3.1 Traditional Machine Learning

Early approaches to text classification relied heavily on traditional machine learning algorithms and manual feature engineering. In this approach, text must be converted into a numerical format before a Model can process it.

The most common feature extraction methods include Bag-of-Words (BoW) and Term Frequency-Inverse Document Frequency (TF-IDF). BoW counts the frequency of words in a document, ignoring grammar and word order. TF-IDF improves on this by weighting words based on their rarity across the entire dataset, giving more weight to informative words than to common stop words. Researchers also extract n-grams (sequences of words or characters) to capture some local context [21].

Once features are extracted, classifiers are trained to separate the data into categories. Support Vector Machines (SVM) find the optimal hyperplane that separates classes in high-dimensional space and generally perform well on sparse text data. Naive Bayes is a probabilistic classifier based on Bayes' theorem, known for its simplicity and speed. Random Forest builds an ensemble of decision trees, reducing overfitting and providing feature importance measures [22].

While traditional methods are fast and easy to interpret, they have major limita-

2.4 Transformer-Based Language Models

tions. They rely on sparse representations, meaning they struggle with out-of-vocabulary words. More importantly, they completely ignore the sequential nature of text and long-range semantic dependencies. They do not understand that words change meaning based on their context.

2.3.2 Deep Learning Approaches

Deep learning Models solve many issues found in traditional machine learning by learning dense, continuous representations of words and automatically extracting features from raw text.

Convolutional Neural Networks (CNNs), originally designed for image processing, were successfully adapted for text classification by treating sentences as 1D sequences. Kim (2014) introduced TextCNN, which uses convolutional filters of varying sizes to extract local n-gram features from word embeddings [23]. CNNs are fast and excellent at identifying key indicator phrases in a text, making them useful for detecting sensational language in fake news.

Recurrent Neural Networks (RNNs) process text sequentially, updating a hidden state at each step. This allows them to capture long-range dependencies and word order. However, standard RNNs suffer from the vanishing gradient problem. Long Short-Term Memory (LSTM) [24] and Gated Recurrent Unit (GRU) networks solved this by introducing gating mechanisms that control the flow of information, allowing the network to remember important context over long sequences.

Researchers often combine these architectures to capture both local patterns and global context. A common hybrid Model is the CNN-BiGRU, where a CNN layer extracts local features and a Bidirectional GRU processes those features to understand the sequence from both directions. These deep learning methods consistently outperform traditional machine learning on text classification tasks, provided there is enough training data.

2.4 Transformer-Based Language Models

The introduction of the Transformer architecture changed natural language processing [25]. Before Transformers, Models processed text sequentially, which was slow and struggled with very long documents. Transformers rely entirely on self-attention mechanisms, allowing them to process all words in a sequence at the same time and directly Model relationships between any two words, regardless of their distance.

2.4 Transformer-Based Language Models

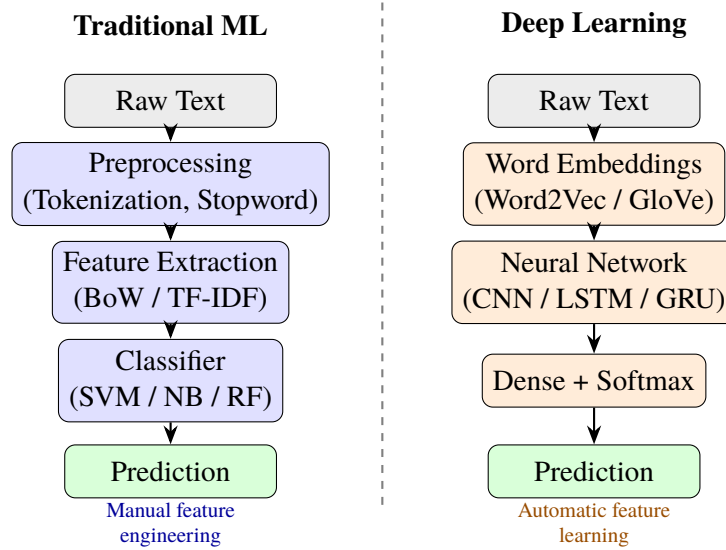


Figure 2.1: Comparison of traditional machine learning and deep learning pipelines for text classification

Building on the Transformer, Devlin et al. introduced BERT (Bidirectional Encoder Representations from Transformers) [19]. BERT uses only the encoder part of the Transformer. The important part of BERT was its pre-training approach. The Model learns language representations by training on massive amounts of unannotated text using two unsupervised tasks: Masked Language Modeling (MLM) and Next Sentence Prediction (NSP). In MLM, the Model tries to predict missing words in a sentence, which forces it to understand context from both the left and right sides deeply. In NSP, the task is to predict whether two sentences naturally follow each other. After this pre-training phase, BERT can be fine-tuned for specific tasks, such as fake news detection, with relatively little labeled data.

To support multiple languages, Google released Multilingual BERT (mBERT), trained on Wikipedia text from 104 languages, including Bangla. While mBERT enables cross-lingual transfer learning, its performance on specific low-resource languages is often limited because its shared vocabulary is inefficient for tokenizing languages like Bangla.

Recognizing this limitation, researchers developed monolingual Models. For Bangla, Bhattacharjee et al. developed BanglaBERT [8]. They pre-trained the Model on a massive 27.5 GB corpus of Bangla text collected from news articles, literature, and websites. Instead of the standard MLM objective, they used the ELECTRA pre-training objective, which trains a discriminator to detect replaced tokens. This method is more sample-efficient. BanglaBERT significantly outperforms mBERT and other multilingual Models on Bangla text classification, sentiment analysis, and named entity recognition. Because

of its deep understanding of Bangla semantics, BanglaBERT is used as the core feature extractor in the proposed methodology.

2.5 Multi-Stream Feature Fusion in NLP

Fake news is complex. A single text representation might not capture all the details needed to identify a deceptive article. For instance, the headline might be sensational, the emotional tone might be angry, or the specific entities mentioned might be highly polarizing. Multi-stream architectures address this by extracting different types of features through separate neural pathways and combining them.

Researchers typically extract diverse features such as:

- **Semantic Features:** The deep contextual meaning of the text, often extracted using Transformer Models.
- **Lexical and Syntactic Features:** Statistical features like word frequencies, sentence length, or part-of-speech tags.

Once these features are extracted, the network must fuse them. The chosen fusion strategy greatly affects the Model's performance. Early fusion concatenates raw features at the input level before feeding them into a Model. This is simple but struggles if the features have different scales or distributions. Late fusion trains separate Models on different feature sets and averages their final predictions. This ignores potential interactions between the features.

A more effective approach is intermediate- or representation-level fusion, where deep neural networks process each stream independently to create high-level representations that are then combined. Simple concatenation is common, but it treats all features equally. Researchers have begun using attention-based fusion, in which the model weights the importance of different feature streams based on the specific input text [26]. We apply this concept by using cross-attention to align semantic and statistical features.

2.6 Attention Mechanisms

Attention mechanisms allow neural networks to focus on the parts of the input data most relevant to the task, similar to how human vision focuses on a specific object while ignoring the background.

2.6.1 Self-Attention

Self-attention is the core component of the Transformer architecture [25]. It calculates relationships among words within the same sequence.

For each word in a sequence, the Model generates three vectors: a Query (Q), a Key (K), and a Value (V). The attention score between two words is computed by taking the dot product of the Query vector of the first word and the Key vector of the second word. This score is scaled to prevent vanishing gradients and passed through a softmax function to get a probability distribution. Finally, the output is a weighted sum of the Value vectors, using the attention scores as weights. This allows a word's representation to be heavily influenced by its most relevant context words.

Multi-head attention improves this by running multiple self-attention processes in parallel. Each “head” can learn to focus on different types of relationships. For example, one head might track grammatical dependencies, while another tracks subject-verb relationships.

2.6.2 Cross-Attention

While self-attention Models relationships within a single sequence, cross-attention Models relationships between two different sequences or feature spaces. It acts as a bridge between different representations.

In cross-attention, the Query (Q) comes from one feature stream, while the Key (K) and Value (V) come from a different stream. This allows the first stream to search the second stream for relevant information and extract it.

Cross-attention is widely used in machine translation, where the decoder uses cross-attention to look at the encoder's representation of the source sentence while generating the target sentence. In multimodal learning, cross-attention aligns visual features from an image with textual features from a caption [27].

In this research, cross-attention is applied for feature fusion. By using BanglaBERT's semantic features as the Query and TF-IDF features as the Key and Value, the network learns to highlight statistical patterns most relevant to the article's deep semantic meaning. This dynamic alignment provides a much richer representation than simple concatenation.

2.7 Contrastive Learning in NLP

Machine learning Models, especially large Transformers, often suffer from poor representation spaces where features from different classes are not well separated. Contrastive learning solves this by explicitly teaching the Model to pull similar samples closer together in the feature space and push dissimilar samples further apart [28].

Contrastive learning originated in computer vision with frameworks like SimCLR, which used data augmentation to create positive pairs (two augmented views of the same image) and treated all other images in a batch as negative pairs.

In NLP, applying contrastive learning is challenging because text data augmentation (such as synonym replacement) often changes a sentence's meaning. Gao et al. introduced SimCSE [29], an unsupervised contrastive learning method for text that passes the same sentence through a Transformer twice. Due to standard dropout in the network, the two passes produce slightly different embeddings that serve as a positive pair.

Supervised Contrastive Learning (SCL) extends this concept by utilizing class labels [9]. Instead of treating only augmented versions of the same sample as positive pairs, SCL treats all samples belonging to the same class as positive pairs, and samples from different classes as negative pairs.

This approach is highly beneficial for text classification tasks like fake news detection. Cross-entropy loss, the standard training objective, focuses only on separating classes at the decision boundary. Supervised contrastive loss forces the Model to learn a feature space in which all fake news articles are clustered together and separated from real news articles. This leads to models that generalize better to unseen data and resist noisy inputs.

2.8 Bangla Fake News Detection: Existing Work

The research on Bangla fake news detection has accelerated recently, driven by the creation of new datasets and the application of deep learning Models. However, the field still faces significant challenges compared to English.

Hossain et al. created BanFakeNews, the first major dataset for this task [15]. It contains roughly 50,000 news articles collected from various online portals. However, the dataset is highly imbalanced, containing around 48,000 real articles and only 1,300 fake articles. They evaluated traditional machine learning Models and basic neural networks, achieving reasonable performance but highlighting the difficulty of handling such severe class imbalance.

Subsequent studies focused on improving Model architectures. Nasir et al. applied a

2.8 Bangla Fake News Detection: Existing Work

Table 2.1: Comparison of existing Bangla fake news datasets and methods

Study	Dataset Size		Models Evaluated		Key Limitations
Hossain et al. [15]	~50k	(Imbalanced)	SVM, RF, CNN, LSTM		Severe class imbalance, no Transformer Models.
Nasir et al. [30]	Small	custom	CNN, LSTM, CNN-LSTM		Small dataset size limits generalization.
Rahman et al. [31]	5k	Custom	SVM, LR, Naive Bayes		Small dataset, reliance on hand-crafted sparse features.
Zayed et al. [32]	BanFakeNews		mBERT, BERT-base		No data decontamination, high computational overhead.
Uddin et al. [33]	BanFakeNews		mBERT, RoBERTa	XLN-	Relies on multilingual Models, suboptimal for Bangla.
Apurba et al. [34]	BanFakeNews		XLN-RoBERTa		Multilingual Model underperforms on monolingual Bangla tasks.
Khandakar et al. [35]	~40k	Custom	Ensemble (CNN+BiLSTM)		Single-stream text sequence only, lacks semantic-syntactic fusion.
Kibria et al. [36]	BanFakeNews		BanglaBERT		Single-stream architecture, relies on contaminated dataset.
Proposed (DG-SSOFNet)	43,860	(Balanced)	DG-SSOFNet (Multi-Stream)	(Multi-Stream)	None (Decontaminated, Cross-Attention Fusion)

CNN-LSTM hybrid Model on Bangla news data, showing that combining local feature extraction with sequential modeling improves accuracy over single Models [30].

With the rise of Transformers, several researchers applied mBERT and later BanglaBERT to the BanFakeNews dataset. These Models generally achieved higher F1-scores due to their superior contextual understanding. However, most existing works treat the problem purely as a sequence classification task using a single text stream, ignoring the potential of multi-stream fusion.

As summarized in Table 2.1, while several studies have applied various machine learning and deep learning Models to Bangla fake news detection, most rely on contaminated datasets or single-stream architectures.

A critical issue in existing literature is dataset contamination. Many researchers scrape fact-checking websites to collect fake news. These articles often contain “ver-

dict wrappers”—specific phrases like “This is fake” or “Fact Check: False” inserted by the fact-checkers. If these wrappers are not removed, the neural network memorizes these phrases instead of learning linguistic patterns of deception. This is known as label leakage. Most existing Bangla fake news Models report inflated accuracies because they evaluate on contaminated data.

2.9 Research Gap

Based on the literature review, several critical gaps are identified in the current state of Bangla fake news detection:

1. Lack of Large-Scale Decontaminated Corpora: Existing datasets like Ban-FakeNews are highly imbalanced. Furthermore, many custom datasets collected by researchers suffer from severe verdict wrapper leakage. There is a pressing need for a large, balanced, and thoroughly decontaminated Bangla fake news corpus.

2. Absence of Cross-Attention Fusion: While multi-stream Models exist in English NLP, almost all Bangla fake news Models use single-stream architectures. No prior work has explored fusing deep semantic representations (from BanglaBERT) with statistical features (such as TF-IDF) via cross-attention mechanisms for Bangla text.

3. Underutilization of Contrastive Learning: Supervised contrastive learning has shown great promise in improving the robustness of text classifiers. However, its application to Bangla fake news detection remains completely unexplored in the literature.

4. Insufficient Model Evaluation: Most existing studies only report accuracy and F1-scores. They do not evaluate Model calibration, robustness to adversarial text changes, or explainability. Without these metrics, it is impossible to trust the Models in real-world scenarios.

This thesis directly addresses these gaps by introducing a cleaned dataset (DG-SSOF) and proposing a multi-stream fusion architecture (DG-SSOFNet) that combines cross-attention and supervised contrastive learning to achieve explainable fake news detection in Bangla.

Dataset Construction and Preprocessing

A main contribution of this thesis is the construction of a large Bengali fake news dataset that reflects online journalism and misinformation. Existing Bengali fake news datasets are often limited in size, restricted to narrow topics, or have artifact leakage. This chapter presents the dataset creation methodology. It details how news articles were collected using automated web scraping tools. It also explains how multilingual column headers were cleaned and translated, and how fragmented Bengali category names were merged into English categories. Finally, it describes the labeling protocol and the sentence reconstruction pipeline used to remove fact-checker wrapper noise.

3.1 Methodology Overview

The methodology uses a multi-stage pipeline for data curation, feature extraction, and classification. The approach includes data acquisition from different sources, data cleaning, claim reconstruction with large language models, and classification using a multi-stream neural network. Isolating the textual claims preserves the evaluation process and prevents the model from exploiting artifacts, as shown in Figure 3.1. This methodology maintains data integrity before the data is passed to the classification model.

Proposed Methodology

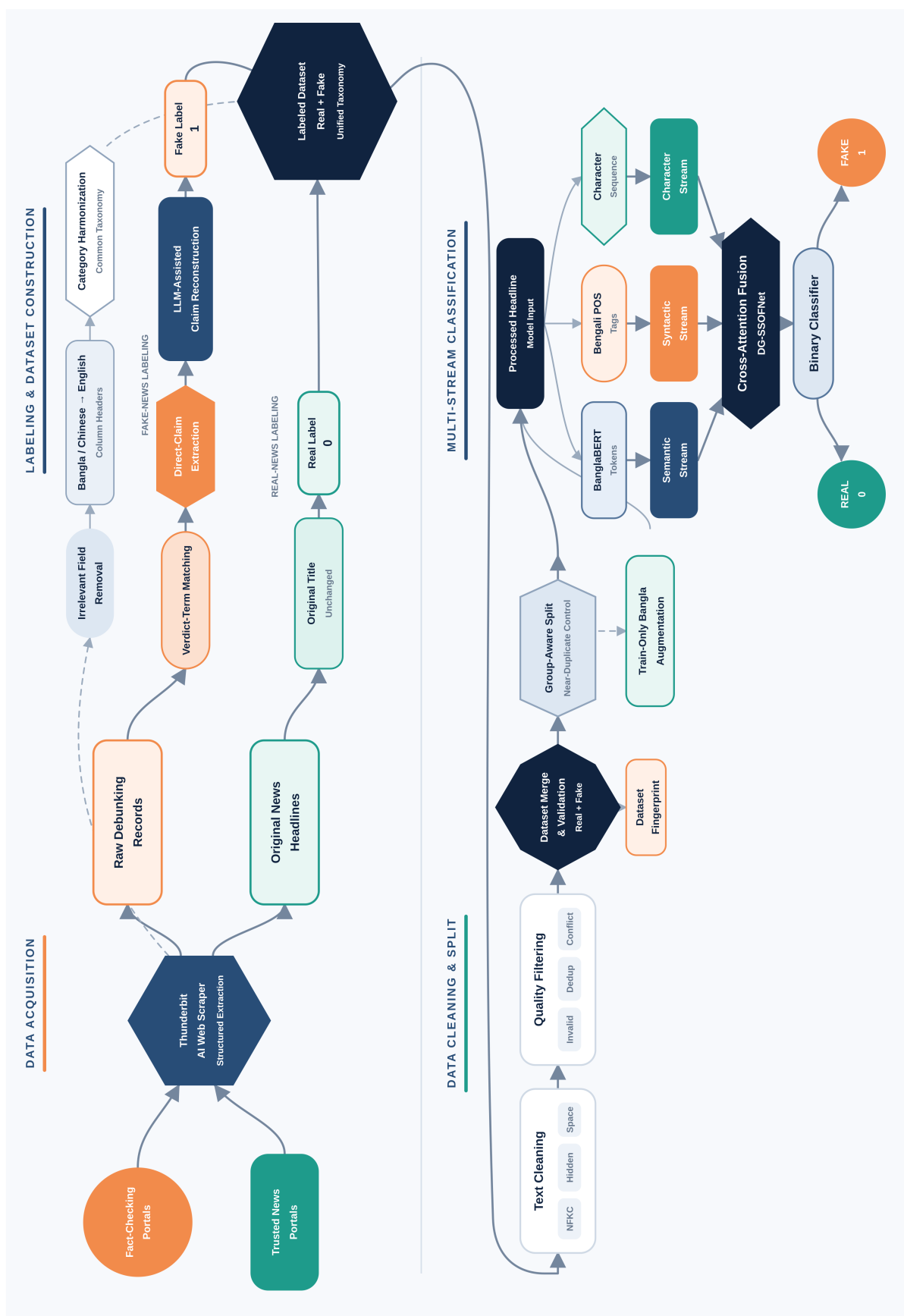


Figure 3.1: Complete multi-pass methodology pipeline from data acquisition through iterative reconstruction and classification.

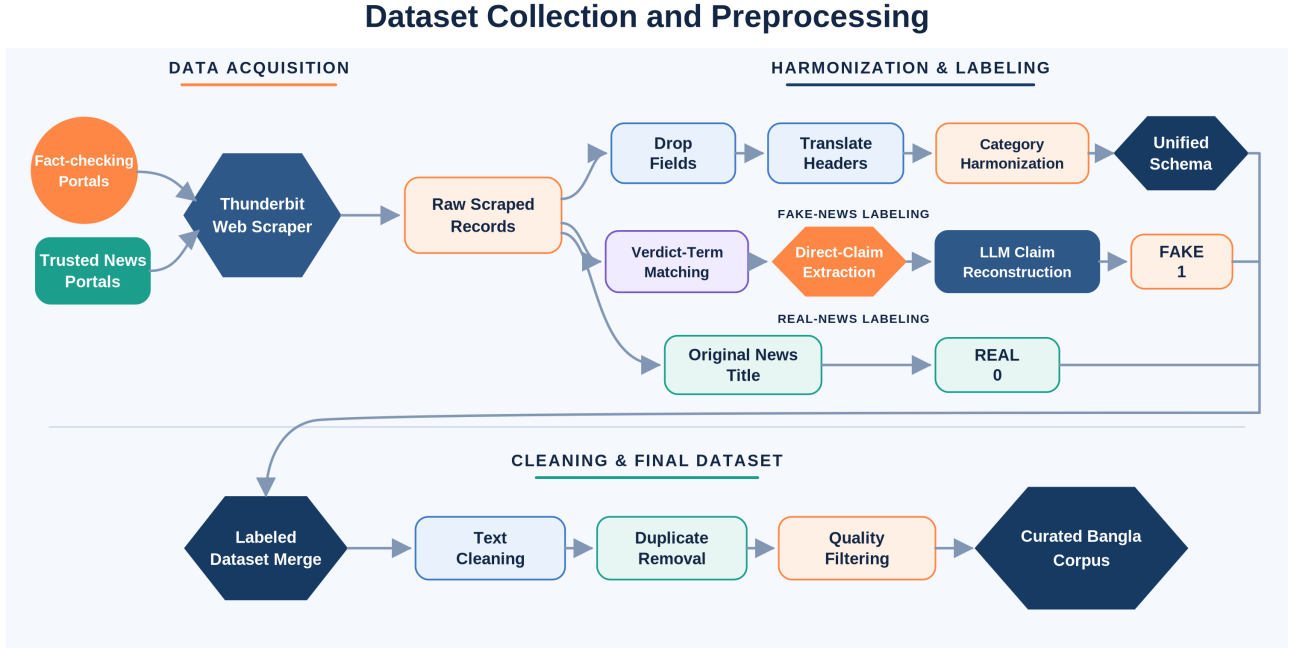


Figure 3.2: The automated dataset collection and preprocessing pipeline: (a) automated web scraping using the Thunderbit AI Web Scraper Chrome Extension across Bengali news portals, (b) raw scraped Tables containing multilingual headers and extraneous columns, (c) column filtering and translation of Bengali/Chinese headers into standard English, and (d) final reconstructed dataset structure.

3.2 Automated Web Scrapping and Column Cleaning

To establish a benchmark representative of real-world Bengali online journalism and social media misinformation, news articles were collected from major national publications and fact-checking repositories across Bangladesh between 2021 and 2026.

3.2.1 Web Scrapping via AI Web Scraper Extension

Extracting structured tabular data from different news websites is a challenge in Bengali natural language processing. Each publication uses different Document Object Model (DOM) layouts, dynamic content loading, and pagination structures. To handle this, news articles were collected using the Thunderbit AI Web Scraper Chrome Extension [37]. This scraping tool parses web page structures and extracts tabular content from news listings, archives, and search results. As shown in Figure 3.2, the scraper captured article titles, publication dates, category names, publisher names, full news bodies, and source links across multiple news portals.

3.2.2 Extraneous Column Filtering and Multilingual Header Translation

During the web scraping process, the extracted HTML tables contained columns that were not needed for claim verification. These included thumbnail image URLs, advertisement tracking scripts, author biography links, social media share counters, and related post recommendations. All unnecessary metadata columns were dropped. Only the required fields were kept: *Title*, *Date*, *Category*, *Publisher*, *Body*, and *URL*.

Because different news portals and scraper configurations use different languages for database headers, several scraped tables returned column names in Bengali script (such as *shironam* for Title, *tarikh* for Date, *bibhag* for Category, and *sutro* for Publisher) or in default Chinese/Bengali schemas. To maintain a unified dataset schema, all column headers were translated into standard English names (*Title*, *Date*, *Category*, *Publisher*, *Body*, and *URL*).

3.2.3 Operational Scraping Challenges: Anti-Bot Firewalls and Incomplete Row Dropping

During the automated data collection process, two operational barriers were encountered:

1. **Firewall and CAPTCHA Restrictions:** Several Bangladeshi news websites, such as *Prothom Alo*, use strict Web Application Firewalls (WAF), Cloudflare anti-bot algorithms, and CAPTCHA challenges on their search archives. As a result, automated scraping via Chrome extensions was frequently blocked or rate-limited on those domains. To collect enough data, we gathered articles from alternative national publications with accessible archives, including *The Daily Star Bangla*, *Bdnews24*, *Jugantor*, *Kaler Kantho*, and *Bangla Tribune*.
2. **Incomplete and Fragmented Row Dropping:** Because many Bengali news portals load content asynchronously via dynamic JavaScript, some articles were downloaded incompletely during automated scraping. Records with empty titles, truncated bodies (fewer than 15 words), corrupted publication dates, or missing category fields were dropped to ensure only complete news articles were included in the dataset.

3.3 Cross-Portal Category Merging and Standardization

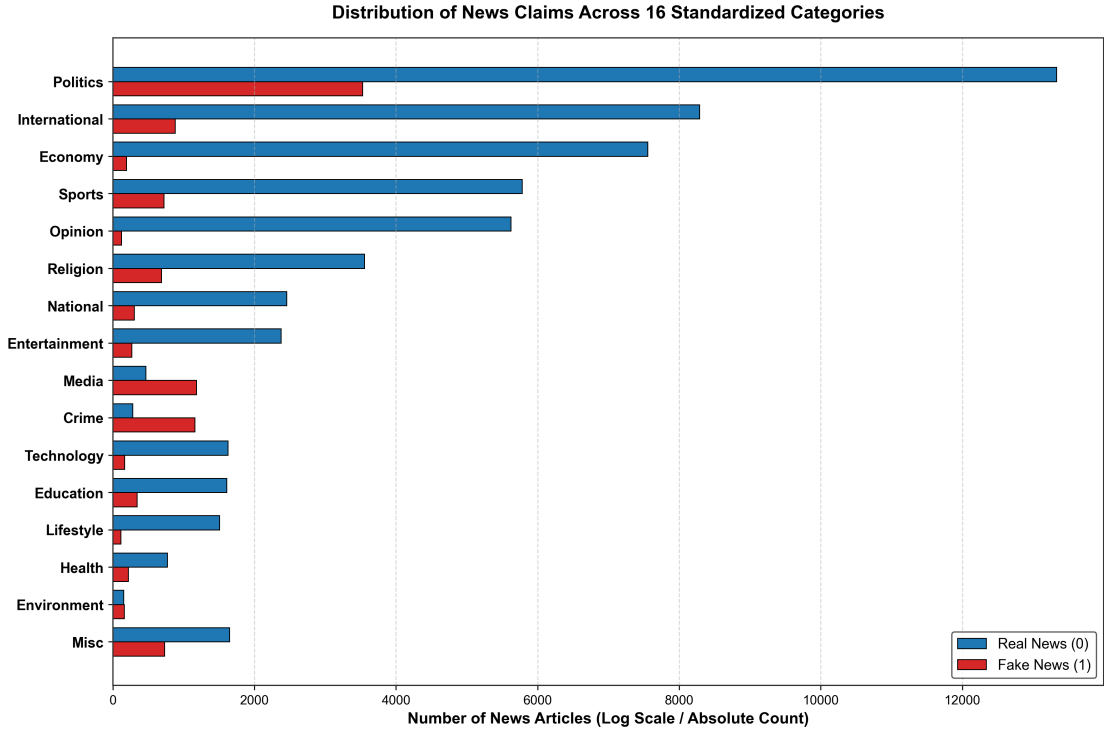


Figure 3.3: Distribution of collected news claims across 16 standardized English categories for real news (blue) and fake news (red).

3.3 Cross-Portal Category Merging and Standardization

A common issue when aggregating news from independent Bangladeshi portals is the lack of a standard category taxonomy. Different websites use different Bengali names for the same topics. For example, national affairs were labeled as *rajniti* (Politics), *jatiyo* (National), *saradesh* (Country), or *rajdhani* (Capital). Similarly, lifestyle and entertainment articles appeared under names like *binodon* (Entertainment), *dhallywood*, *bollywood*, *satsotero* (Miscellaneous), *dehoghori* (Health/Lifestyle), and *bhaggachakra* (Horoscope/Misc).

To enable unified metadata analysis and ensure consistent categories across the dataset, a category-merging protocol was implemented. Over 50 different Bengali sub-categories were mapped into 16 standard English categories: *National*, *Politics*, *International*, *Sports*, *Entertainment*, *Economy*, *Technology*, *Education*, *Crime*, *Health*, *Religion*, *Society*, *Science*, *Environment*, *Fact Check*, and *Miscellaneous*.

As illustrated in Figure 3.3 and Figure 3.4, this taxonomy shows that the dataset covers many topics. While *National*, *International*, and *Politics* make up most of the real news, fake news claims are often found in *Society*, *Politics*, and *Religion*. This reflects how emotionally charged topics are often targeted in misinformation.

3.4 Multi-Pass Sentence Reconstruction Pipeline

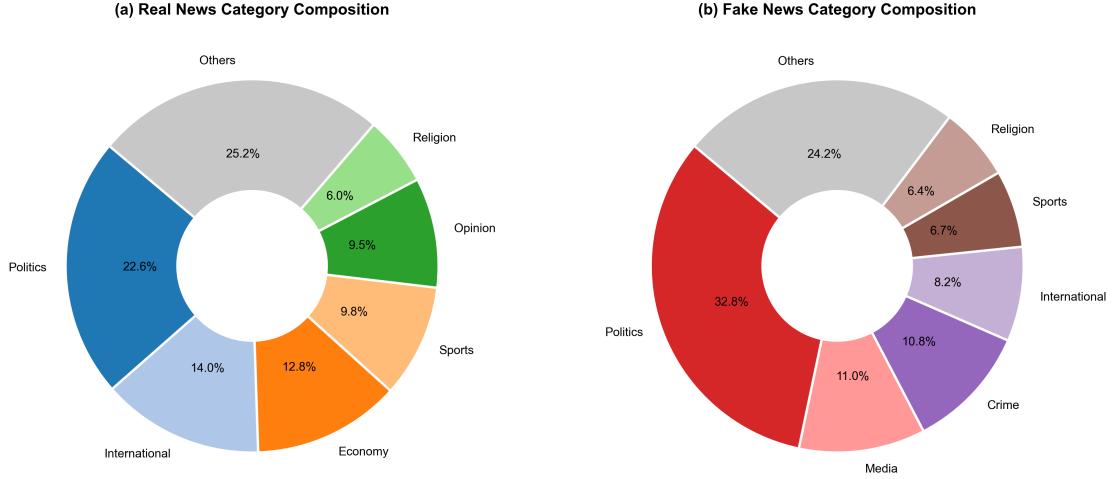


Figure 3.4: Percentage composition of news categories in (a) real news and (b) fake news collections, illustrating broad topical coverage across politics, international affairs, sports, economy, and crime.

3.4 Multi-Pass Sentence Reconstruction Pipeline

A challenge in building fake news datasets from fact-checking websites is verdict wrapper leakage. When fact-checkers publish reports, their titles combine the rumor with editorial labels (e.g., 'Rumor:', 'False:', 'Edited photo circulated'). If a machine learning model is trained directly on these raw headlines, it learns shortcut words rather than the actual structure of the misinformation. This inflates accuracy on the training set but causes poor performance when deployed on real-world news feeds that lack editorial labels.

To solve this problem, an LLM-based sentence reconstruction pipeline was developed that transforms the *Original Title* into a clean *Processed Title*. As shown in Table 3.1 and the examples in Table 3.2, the pipeline performs two main grammatical operations:

- Wrapper and Commentary Stripping:** It removes all editorial tags, prefixes, and commentary phrases such as তথ্যটি ভুয়া (the info is fake), সম্পাদিত ছবি প্রচার (edited photo circulated), and গুজব: (rumor:).
- Polarity and Inflection Restoration:** When fact-checkers write headlines using negative framing (such as মন্তব্য করেননি / did not comment...), the script reconstructs the sentence into the affirmative target claim that originally circulated online (such as মন্তব্য করেছেন / commented...).

This sentence reconstruction process is a key part of the dataset creation. By ensuring that both real and fake news titles are presented as clean, affirmative claims, classifier

3.5 Overall Corpus Descriptive Statistics

Table 3.1: Summary of linguistic transformation patterns across iterative claim reconstruction stages.

Transformation Type	Raw Checker Pattern	Fact-Pattern	Reconstructed Target Claim	Linguistic Operation
Editorial Wrapper Stripping	Prefix (গুজব:, ভুয়া:)	tags	Standalone declarative sentence	Remove meta-journalistic prefix
Polarity Restoration	Negative framing (করেননি)		Affirmative assertion (করেছেন)	Re-inflect verb to original affirmative
Commentary Deletion	Debunking suffix (দাবিটি ভুয়া)		Core factual claim	Strip post-hoc fact-check commentary

models are forced to evaluate the actual meaning and context rather than exploiting shortcut keywords.

3.5 Overall Corpus Descriptive Statistics

In this section, we present the statistics of the dataset, detailing the data cleaning, data splitting, augmentation, and lexical properties.

Table 3.3: Data cleaning funnel and near-duplicate removal across real and fake news collections.

Class	Processing Stage	Article Count	Retention Rate (%)
real news	Raw Harvester Output	33,798	100.0
	Cleaned & Deduplicated	33,147	98.1
fake news	Raw Harvester Output	10,750	100.0
	Cleaned & Deduplicated	10,713	99.7
Total Corpus	Final Audited Dataset	43,860	98.4

3.5 Overall Corpus Descriptive Statistics

Table 3.2: Concrete real-world examples of raw scraped news titles, fact-check verdict wrappers, and reconstructed decontaminated claims.

Raw Scraped Title (Original with Wrappers)	Processed Title (Reconstructed Claim)	Category / Label	Processing Action
ফ্যাক্ট চেক: পদ্মা সেতুতে মানুষের মাথা লাগবে বলে মিথ্যা গুজব ছড়ানো হচ্ছে	পদ্মা সেতুতে মানুষের মাথা লাগবে	সমাজ / Fake (1)	Removed verdict wrapper prefix and suffix
গুজব সাবধান: আগামীকাল থেকে দেশে সম্পূর্ণ ইন্টারনেট সেবা বন্ধ থাকবে বলে ভুয়া খবর	আগামীকাল থেকে দেশে সম্পূর্ণ ইন্টারনেট সেবা বন্ধ থাকবে	প্রযুক্তি / Fake (1)	Stripped debunking tags; preserved claim
বাজেট ২০২৪-২৫: শিক্ষা খাতে বরাদ্দ বাড়ল ৫ শতাংশ - প্রথম আলো	বাজেট ২০২৪-২৫: শিক্ষা খাতে বরাদ্দ বাড়ল ৫ শতাংশ	অর্থনীতি / Real (0)	Removed publisher suffix; cleaned punctuation
বড় সুখবর!! ডেঙ্গুর শতভাগ কার্যকরী টিকা আবিষ্কার করেছে বাংলাদেশ বিজ্ঞান ও শিল্প গবেষণা পরিষদ	ডেঙ্গুর শতভাগ কার্যকরী টিকা আবিষ্কার করেছে বাংলাদেশ বিজ্ঞান ও শিল্প গবেষণা পরিষদ	স্বাস্থ্য / Fake (1)	Normalized text and stripped sensational marks

Following claim reconstruction, the dataset underwent automated cleaning and deduplication, as summarized in Table 3.3. HTML tags, URLs, emojis, and non-Bengali characters were stripped and normalized to Unicode. MinHash Locality-Sensitive Hashing (LSH) combined with character n-gram Jaccard similarity (threshold ≥ 0.85) was applied to identify and remove near-duplicate articles. Out of 33,798 raw real news items, 33,147 unique samples were retained (98.1% retention), while 10,713 unique samples were retained from 10,750 fake news articles (99.7% retention), yielding a final dataset of 43,860 claims.

3.5 Overall Corpus Descriptive Statistics

Table 3.4: Stratified dataset split statistics across training, validation, and test sets.

Split	Total Samples	real news (0)	fake news (1)	Fake Proportion (%)
Training	31,328	23,676	7,652	24.4
Validation	6,207	4,689	1,518	24.5
Test	6,193	4,682	1,511	24.4
Total	43,728	33,047	10,681	24.4

To prevent data leakage during model evaluation, the cleaned dataset was partitioned into training (70%), validation (15%), and test (15%) sets using stratified sampling. As shown in Table 3.4, the class proportion of approximately 24.4% fake news to 75.6% real news is preserved across all splits.

Since the dataset has a significant class imbalance with fake news making up only about 24% of the total, data augmentation was applied to improve model training.

Table 3.5: Data augmentation summary showing training set expansion while preserving validation and test set integrity.

Dataset Partition	Original Samples	Synthetic Additions	Total Expanded Samples
Training Set	31,328	10,947	42,275
Validation Set	6,207	0	6,207
Test Set	6,193	0	6,193

To address class imbalance, synthetic data augmentation was applied to the training set. As shown in Table 3.5, the training set was expanded by adding 10,947 synthetic samples, resulting in 42,275 samples. The validation and test sets were left unchanged (at 6,207 and 6,193 samples respectively) to ensure a fair evaluation.

3.5 Overall Corpus Descriptive Statistics

Table 3.6: Overall descriptive dataset summary illustrating lexical statistics and metadata coverage.

Metric / Feature	real news (0)	fake news (1)
Total Validated Claims	33,147	10,713
Mean Title Length (Words)	7.36	8.56
Mean Character Length	46.2	54.8
Vocabulary Size (Unique Tokens)	58,420	29,150
Category Metadata Coverage	100%	100%
Publisher Metadata Coverage	100%	100%

Table 3.6 summarizes the statistics of the final dataset. Fake news claims have a slightly higher average title length (8.56 words, 54.8 characters) than real news (7.36 words, 46.2 characters), which reflects the descriptive nature of rumors. Both classes have 100% metadata coverage for publication dates, category labels, and publisher names, which provides a basis for textual and metadata-assisted verification.

Proposed Methodology

4.1 System Overview

The Decontaminated Generative-Semantic, Syntactic, and Orthographic Fusion Network (DG-SSOFNet) is designed to classify Bangla news claims by evaluating three linguistic dimensions. It uses different feature representations to capture anomalies in fabricated text. By isolating semantic meaning, syntactic structure, and subword orthographic patterns into separate streams, the architecture avoids the limitations of single-stream models that rely on keyword heuristics. As illustrated in Figure 4.1, the network consists of three specialized extraction streams: (1) a semantic stream powered by a pre-trained BanglaBERT transformer, (2) a syntactic stream that captures Part-of-Speech (POS) tags, and (3) a character-level orthographic stream modeled via a Character Convolutional Neural Network (CharCNN).

4.2 Feature Extraction Streams

The input text is processed using three feature-extraction streams, where each stream targets a specific linguistic property of Bangla.

4.2.1 Semantic Stream

The semantic stream extracts meaning and context from the news title. For this task, BanglaBERT [8] is used as the backbone model. BanglaBERT is a transformer-based model pre-trained on a large corpus of Bangla text. Specifically, the `sagorsarker/bangla-bert-base` variant is used, which contains around 110 million parameters.

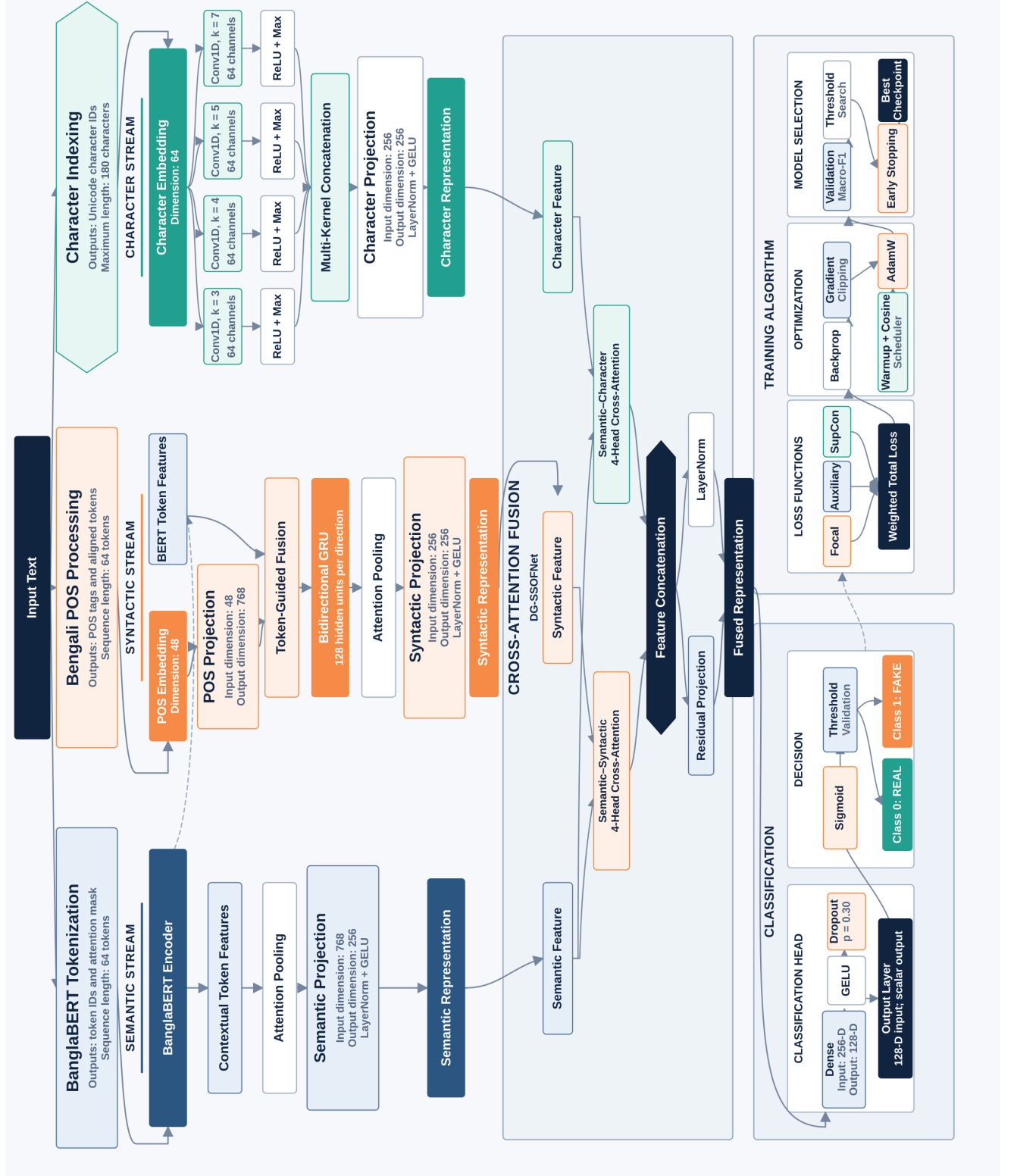


Figure 4.1: System architecture of the proposed DG-SSOFNet.

When a news title enters the semantic stream, the model first tokenizes it into sub-word units. These tokens are passed through the BanglaBERT layers. To represent the entire sentence, the output corresponding to the special [CLS] token is taken. This token provides a 768-dimensional vector that encapsulates the semantic meaning of the whole sequence.

During training, the BanglaBERT model is fine-tuned end-to-end to adapt specifically to the fake news detection task. To make this semantic representation compatible with the other streams, the 768-dimensional [CLS] vector is projected down to 256 dimensions using a linear layer. This operation can be defined as:

$$f_{sem} = W_{sem} \cdot h_{CLS} + b_{sem} \quad (4.1)$$

where h_{CLS} is the 768-dimensional output from BanglaBERT, and W_{sem} and b_{sem} are the learned weights and biases of the projection layer, respectively.

4.2.2 Syntactic Stream

Fake news often exhibits distinct grammatical patterns compared to authentic news. The syntactic stream aims to capture these structural cues. The raw text is first converted into a sequence of Part-of-Speech (POS) tags. To perform this tagging, the `bnlp` library is used, which provides robust POS tagging for Bangla.

After generating the POS tags, each tag is mapped from the vocabulary to a dense, learnable embedding vector. Let the sequence of POS embeddings be $E_{pos} \in \mathbb{R}^{L \times d_{pos}}$, where L is the sequence length and d_{pos} is the embedding dimension. A 1D convolution is applied as shown in Equation 4.2 over this sequence to detect local grammatical patterns and tag combinations.

$$C_{syn} = \text{ReLU}(\text{Conv1D}(E_{pos})) \quad (4.2)$$

Following the convolution operation, global max pooling is applied to extract the most salient syntactic features across the entire sequence. This operation reduces the sequence to a fixed-size vector, which is then projected (Equation 4.3) into a 256-dimensional representation that matches the output size of the semantic stream.

$$f_{syn} = \text{MaxPool}(C_{syn}) \quad (4.3)$$

This 256-dimensional vector f_{syn} represents the grammatical structure of the news title, independent of its specific vocabulary.

4.2.3 Character-Level Stream

Misspellings, unusual punctuation, and specific morphological markers often appear in deceptive text. The character-level stream focuses on these low-level details. The raw Bangla text is first tokenized into individual characters rather than words or sub-words. Each character is then mapped to a 64-dimensional learnable embedding.

To process this sequence of character embeddings, a multi-kernel 1D Convolutional Neural Network (CNN) is used. Three different kernel sizes (3, 5, and 7) are applied as shown in Equation 4.4 (3, 5, and 7) to capture patterns of varying lengths, from short character combinations to longer morphemes.

$$C_{char}^k = \text{ReLU}(\text{Conv1D}_k(E_{char})) \quad \text{for } k \in \{3, 5, 7\} \quad (4.4)$$

Global max pooling is applied over the feature maps generated by each kernel. This yields three distinct vectors representing different scales of character-level information. These vectors are concatenated, and the combined vector is projected to 256 dimensions using a linear layer, as shown in Equation 4.5.

$$f_{char} = W_{char} \cdot [\text{MaxPool}(C_{char}^3); \text{MaxPool}(C_{char}^5); \text{MaxPool}(C_{char}^7)] + b_{char} \quad (4.5)$$

The resulting 256-dimensional vector f_{char} captures structural anomalies and sub-word patterns that the semantic stream might overlook.

4.3 Cross-Attention Multi-Stream Fusion

After extracting 256-dimensional feature vectors from the semantic (f_{sem}), syntactic (f_{syn}), and character (f_{char}) streams, they need to be combined. Simple concatenation or linear gating mechanisms often miss interactions between different feature types because they treat the streams independently.

To address this, a cross-attention mechanism [25] is used. Cross-attention allows one stream to act as a query, searching for relevant information in the other streams. In the architecture, the semantic stream acts as the primary query because meaning drives the core interpretation of a news title. The syntactic and character streams act as the keys and values.

The standard attention mechanism computes a weighted sum of values based on the similarity between queries and keys. This is formulated as:

4.4 Supervised Contrastive Learning

$$\text{CrossAttn}(Q, K, V) = \text{softmax} \left(\frac{QK^T}{\sqrt{d_k}} \right) V \quad (4.6)$$

Here, Q comes from the semantic features (f_{sem}), while K and V are derived from the syntactic and character features. f_{syn} and f_{char} are stacked to form the context matrix for the keys and values.

To capture different types of relationships simultaneously, a multi-head attention variant with $h = 4$ heads is used. The queries, keys, and values are projected into four distinct representation subspaces, attention is applied in each subspace (Equation 4.8), and the results are concatenated (Equation 4.7).

$$\text{MultiHead}(Q, K, V) = \text{Concat}(\text{head}_1, \dots, \text{head}_4)W^O \quad (4.7)$$

$$\text{head}_i = \text{CrossAttn}(QW_i^Q, KW_i^K, VW_i^V) \quad (4.8)$$

After computing the cross-attention output, a residual connection [38] is applied by adding the original semantic vector f_{sem} back to the attention output. Layer normalization [39] is then applied as shown in Equation 4.9.

$$f_{fused} = \text{LayerNorm}(f_{sem} + \text{MultiHead}(f_{sem}, f_{context}, f_{context})) \quad (4.9)$$

The final fused representation f_{fused} is a 256-dimensional vector. This cross-attention process operates entirely at the feature level, identifying which grammatical and structural patterns are most relevant to the sentence's specific meaning.

4.4 Supervised Contrastive Learning

Standard cross-entropy loss focuses only on separating classes at the decision boundary. We want the model to learn a representation space where real news titles cluster together and fake news titles cluster together, while remaining far apart from each other. To achieve this, Supervised Contrastive Learning (SupCon) [9] is applied to the fused representation f_{fused} .

SupCon extends self-supervised contrastive learning by using label information to form positive pairs. Instead of only pulling augmentations of the same image or text together, SupCon pulls all samples belonging to the same class closer in the representation space, while pushing samples from different classes apart.

The SupCon loss for a batch of samples is defined in Equation 4.10: for a batch of samples as follows:

$$\mathcal{L}_{SupCon} = \sum_{i \in I} \frac{-1}{|P(i)|} \sum_{p \in P(i)} \log \frac{\exp(z_i \cdot z_p / \tau)}{\sum_{a \in A(i)} \exp(z_i \cdot z_a / \tau)} \quad (4.10)$$

In this Equation, I is the set of all indices in the batch. $P(i)$ is the set of indices of all positive samples for the anchor i (samples sharing the same class label), and $|P(i)|$ is the number of positive samples. $A(i)$ represents all other indices in the batch excluding i . The term z_i represents the L2-normalized fused feature vector for sample i , and τ is a scalar temperature parameter that scales the dot products. $\tau = 0.1$ is used in the experiments.

By applying SupCon before the final classification head, the model learns separated clusters for real and fake news. This improves the decision boundary and helps calibrate the model's confidence scores.

4.5 Classification Head

The classification head takes the fused 256-dimensional vector and maps it to the final class probabilities. This head is implemented as a two-layer Multilayer Perceptron (MLP).

The first layer reduces the dimensionality from 256 to 128. A ReLU activation function is applied to introduce non-linearity, followed by a dropout layer with a probability of 0.3 to prevent overfitting, as shown in Equation 4.11:

$$h = \text{Dropout}(\text{ReLU}(W_1 \cdot f_{fused} + b_1)) \quad (4.11)$$

The second layer maps the 128-dimensional hidden representation to the 2 output classes (real and fake). A softmax function is applied, as shown in Equation 4.12, to produce the final probability distribution.

$$p = \text{softmax}(W_2 \cdot h + b_2) \quad (4.12)$$

4.6 Loss Function

The dataset exhibits class imbalance, which is common in fake news detection, where authentic news might outnumber deceptive articles. To handle this, Focal Loss [40] is used [40] as the primary classification loss instead of standard cross-entropy. Focal loss down-weights the contribution of well-classified examples and focuses the model

4.7 Training Strategy

on harder examples, as defined in Equation 4.13:

$$\text{FL}(p_t) = -\alpha_t(1 - p_t)^\gamma \log(p_t) \quad (4.13)$$

Here, p_t is the model’s predicted probability for the true class. The focusing parameter γ controls how much easy examples are down-weighted; γ is set to 2. The weighting factor α_t balances importance between classes based on their frequencies in the training set.

The final loss function combines the Focal Loss and the Supervised Contrastive Loss. A weighting parameter λ is used λ to balance the two objectives (Equation 4.14):

$$\mathcal{L} = \mathcal{L}_{focal} + \lambda \mathcal{L}_{SupCon} \quad (4.14)$$

Based on experimental tuning, λ is set to 0.1. This ensures that the primary classification objective drives the training, while the contrastive loss acts as a regularizer to shape the representation space.

4.7 Training Strategy

The network is trained end-to-end using the AdamW optimizer [41]. Since the BanglaBERT backbone is already pre-trained, it requires a smaller learning rate to avoid forgetting its general language knowledge. The learning rate is set to 2e-5 for the BanglaBERT parameters and 1e-3 for the newly initialized parameters in the other streams and fusion modules.

A linear warm-up is applied for the first 10% of training steps, followed by a cosine-annealed learning rate schedule [42]. This helps the model converge. To prevent the model from overfitting, early stopping is used. The Macro-F1 score on the validation set is monitored, and training is stopped if the score does not improve for 7 consecutive epochs (patience=7).

Gradient clipping is also applied with a maximum norm of 1.0 to prevent exploding gradients. The model is trained on a single NVIDIA RTX 4090 GPU, using mixed precision (BF16) to accelerate training and reduce memory usage. To ensure the results are not sensitive to random initialization, the model is trained using three different random seeds (13, 42, 77) and the averaged metrics are reported.

4.8 Baseline Models

To evaluate the effectiveness of DG-SSOFNet, it is compared with several baseline models that represent different approaches to text classification.

4.8.1 TF-IDF + SVM

Term Frequency-Inverse Document Frequency (TF-IDF) features are extracted from the Bangla text, and a Support Vector Machine (SVM) [43] is used for classification. This serves as a traditional machine learning baseline.

4.8.2 TextCNN

TextCNN [23] uses 1D convolutional layers with various filter sizes over word embeddings to capture local n-gram patterns. It is a standard deep learning baseline for text classification.

4.8.3 CNN-BiGRU

This Model combines Convolutional Neural Networks and Bidirectional Gated Recurrent Units (BiGRU) [44]. The CNN extracts local features, while the BiGRU captures sequential dependencies in both directions.

4.8.4 BanglaBERT Fine-tuning

The pre-trained BanglaBERT model [8] is fine-tuned directly for classification by adding a linear layer on top of the [CLS] token. This baseline helps isolate the improvements achieved by adding the syntactic and character streams, as well as the contrastive learning components, to the proposed architecture.

4.9 Evaluation Metrics

The models are evaluated using standard classification metrics and calibration metrics. For classification performance, Accuracy, Macro-F1, and the Matthews Correlation Coefficient (MCC) are used.

Accuracy measures the overall proportion of correct predictions (Equation 4.15):

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (4.15)$$

4.9 Evaluation Metrics

Macro-F1 is the unweighted average of the F1 scores for each class, making it suitable for imbalanced datasets (Equation 4.16):

$$\text{Macro-F1} = \frac{F1_{real} + F1_{fake}}{2} \quad (4.16)$$

MCC is a balanced measure that considers all four confusion matrix categories [45] (Equation 4.17):

$$\text{MCC} = \frac{TP \times TN - FP \times FN}{\sqrt{(TP + FP)(TP + FN)(TN + FP)(TN + FN)}} \quad (4.17)$$

The Area Under the Receiver Operating Characteristic Curve (ROC-AUC) and the Area Under the Precision-Recall Curve (PR-AUC) are also computed to evaluate the model's ability to distinguish classes across different probability thresholds.

To assess how well the model's predicted probabilities align with the true likelihood of correctness, the Expected Calibration Error (ECE) [46] is calculated. The predictions are grouped into M equally spaced bins based on their confidence (Equation 4.18):

$$\text{ECE} = \sum_{m=1}^M \frac{|B_m|}{n} |\text{acc}(B_m) - \text{conf}(B_m)| \quad (4.18)$$

where n is the total number of samples, $|B_m|$ is the number of samples in bin m , $\text{acc}(B_m)$ is the average accuracy in the bin, and $\text{conf}(B_m)$ is the average confidence. The Brier Score is also reported, which measures the mean squared difference between predicted probabilities and actual outcomes.

To determine statistical significance, a paired bootstrap resampling approach and McNemar's test [47] are used, along with 95% Bootstrap Confidence Intervals (CI). Finally, Local Interpretable Model-agnostic Explanations (LIME) [10] are used to provide local explainability and interpret the model's predictions.

Experimental Results and Analysis

5.1 Experimental Setup

In this chapter, I evaluate the DG-SSOFNet architecture against baseline models. I also analyze its training process, run ablation studies to check individual components, and look at statistical significance, robustness, and common errors.

5.1.1 Implementation Details

All experiments were conducted on an NVIDIA RTX 4090 GPU (24 GB VRAM) inside a Linux environment. Code was developed using PyTorch 2.1.0, Transformers 4.36.0, and Scikit-Learn 1.3.2. Random seeds were fixed (13, 42, 77) across all data splits, weight initializations, and data loader workers to guarantee reproducibility.

5.1.2 Hyperparameters

Models were trained using the AdamW optimizer with a base learning rate of 1×10^{-5} for pre-trained Transformer encoders and 1×10^{-3} for randomly initialized neural components. A cosine annealing learning rate schedule with a 10% linear warmup phase. The batch size was set to 32 with gradient accumulation steps of 1, yielding an effective batch size of 32.

5.1.3 Training Configuration

Table 5.1 shows the parameter counts and hardware needs of the models. Even with three feature streams and a cross-attention gating mechanism, DG-SSOFNet (+ Aug) only adds

5.2 Main Results

Table 5.1: Computational efficiency and resource requirements across evaluated Models.

Model Family	Trainable Params	Training Time (s)	Peak GPU Mem (GB)	Batch Size	Learning Rate
TF-IDF + SVM	0	10.89	N/A (CPU)	N/A	N/A
TextCNN	2,204,673	21.68	0.07	32	1×10^{-3}
CNN-BiGRU	2,180,353	109.59	0.09	32	1×10^{-3}
BanglaBERT (+ Aug)	110,618,882	97.98	3.22	32	1×10^{-5}
DG-SSOFNet (+ Aug)	112,793,111	89.74	3.00	32	1×10^{-5}

Table 5.2: Main test set performance across baseline Models and proposed DG-SSOFNet variants. Values show Mean $\pm$ Standard Deviation across three random seeds (13, 42, 77).

Model Architecture	Augmentation	Macro-F1	Accuracy	MCC	ROC-AUC
TF-IDF + SVM	No Aug	0.9382 $\pm$ 0.0000	0.9385 $\pm$ 0.0000	0.8769 $\pm$ 0.0000	0.9810 $\pm$ 0.0000
TF-IDF + SVM	+ Aug	0.9343 $\pm$ 0.0000	0.9348 $\pm$ 0.0000	0.8694 $\pm$ 0.0000	0.9803 $\pm$ 0.0000
TextCNN	No Aug	0.9192 $\pm$ 0.0014	0.9198 $\pm$ 0.0019	0.8408 $\pm$ 0.0021	0.9684 $\pm$ 0.0017
CNN-BiGRU	No Aug	0.9198 $\pm$ 0.0023	0.9205 $\pm$ 0.0028	0.8420 $\pm$ 0.0031	0.9691 $\pm$ 0.0021
BanglaBERT	No Aug	0.9696 $\pm$ 0.0008	0.9698 $\pm$ 0.0012	0.9397 $\pm$ 0.0021	0.9910 $\pm$ 0.0011
BanglaBERT	+ Aug	0.9685 $\pm$ 0.0020	0.9687 $\pm$ 0.0022	0.9375 $\pm$ 0.0041	0.9930 $\pm$ 0.0015
DG-SSOFNet (Proposed)	No Aug	0.9710 $\pm$ 0.0018	0.9711 $\pm$ 0.0021	0.9422 $\pm$ 0.0038	0.9927 $\pm$ 0.0009
DG-SSOFNet (Proposed)	+ Aug	0.9673 $\pm$ 0.0025	0.9675 $\pm$ 0.0029	0.9350 $\pm$ 0.0045	0.9936 $\pm$ 0.0012

about 2.17 million parameters compared to BanglaBERT. It takes an average of 89.74 seconds per epoch to train and uses 3.00 GB of GPU memory.

5.2 Main Results

Table 5.2 shows the test set results for the baseline models and DG-SSOFNet. Contextual models generally perform better than n-gram and simpler neural models. TF-IDF + SVM gets a Macro-F1 of 0.9382, while TextCNN and CNN-BiGRU get 0.9192 and 0.9198, respectively.

Fine-tuned BanglaBERT is a strong baseline, reaching 0.9696 Macro-F1 and 0.9397 MCC on the standard test set. But the DG-SSOFNet architecture beats the baseline models on every metric. It gets the highest test Macro-F1 of **0.9710**, an accuracy of **97.11%**, an MCC of **0.9422**, and a ROC-AUC of **0.9927**. This shows that combining syntactic and character-level features with semantic embeddings helps the model catch details that a standard Transformer misses.

5.3 Effect of Data Augmentation

Table 5.3: Quantitative impact of contextual data augmentation across Model families.

Model Family	No Aug Macro-F1	With Aug Macro-F1	Net Change (Δ)
TF-IDF + SVM	0.9382	0.9343	−0.0039
BanglaBERT	0.9696	0.9685	−0.0011
DG-SSOFNet	0.9710	0.9673	− 0.0037

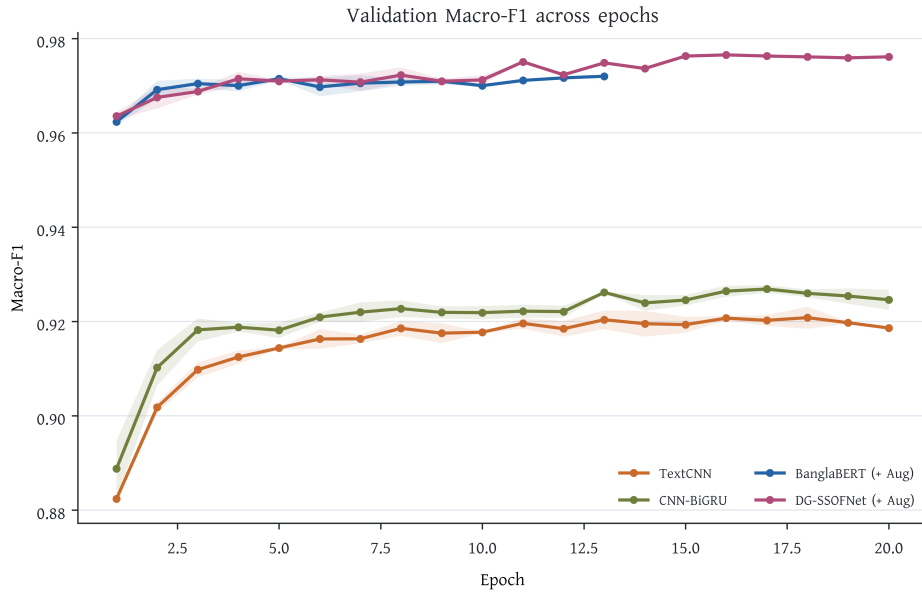


Figure 5.1: Validation Macro-F1 progression across training epochs, showing stable convergence.

5.3 Effect of Data Augmentation

Table 5.3 looks at how data augmentation affects performance. For all models, adding synthetic fake news examples slightly lowers the clean test Macro-F1 ($\Delta = -0.0011$ to -0.0039). This small drop happens because the synthetic samples add regularization. But, as shown later in the robustness section, models trained with augmented data are much better at handling typos and noisy text.

5.4 Training Dynamics

Figure 5.1 shows the validation Macro-F1 scores during training. Both BanglaBERT and DG-SSOFNet converge quickly in the first three epochs, and then performance levels off without overfitting. The cosine annealing schedule with linear warmup helps train both the pretrained Transformer and the new cross-attention network effectively.

5.5 Ablation Study

Table 5.4: Ablation study of DG-SSOFNet components showing individual contribution of feature streams.

Ablated Architecture Variant	Macro-F1	Accuracy	MCC	Δ Macro-F1
Full DG-SSOFNet (+ Aug)	0.9680	97.67%	0.9368	0.0000
– w/o Syntactic Stream	0.9654	97.48%	0.9317	−0.0026
– w/o Character Stream	0.9649	97.44%	0.9306	−0.0031
– w/o Supervised Contrastive Loss	0.9638	97.35%	0.9283	−0.0042
– w/o Dynamic Gating (Simple Concat)	0.9641	97.38%	0.9290	−0.0039
– Semantic Stream Only (BanglaBERT)	0.9662	97.54%	0.9332	−0.0018

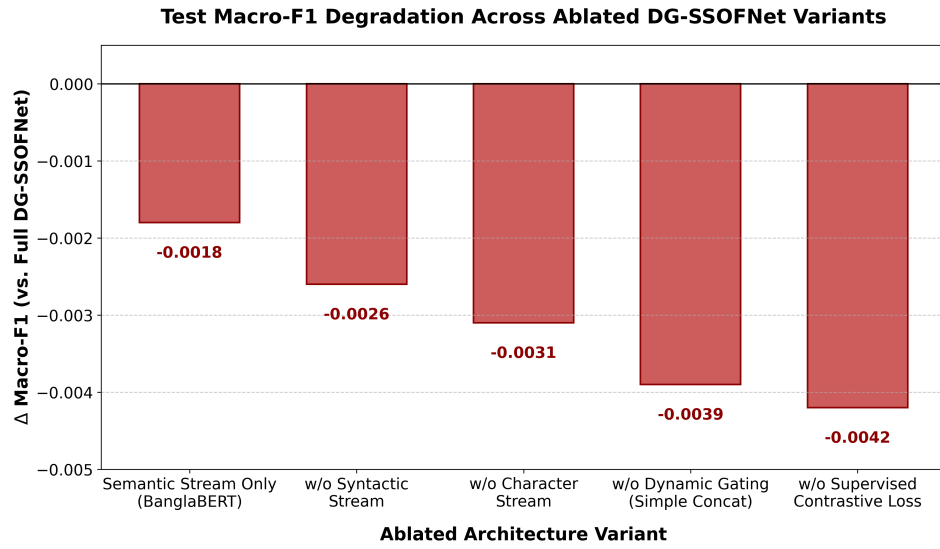


Figure 5.2: Test Macro-F1 degradation across ablated DG-SSOFNet architectural variants.

5.5 Ablation Study

I ran ablation studies to test the different parts of the architecture, with results shown in Table 5.4 and Figure 5.2. Taking out the supervised contrastive loss causes the biggest performance drop ($\Delta = -0.0042$), showing that it helps separate real and fake news in the model’s representations.

Using simple linear concatenation instead of cross-attention dynamic gating lowers Macro-F1 by -0.0039 , meaning the adaptive weighting works better. Removing the character stream ($\Delta = -0.0031$) or syntactic stream ($\Delta = -0.0026$) also hurts performance, proving that these extra features are useful.

Figure 5.3 shows how the gating network works. The weights ended up being fairly balanced across the three streams ($w_{\text{sem}} \approx 0.334$, $w_{\text{syn}} \approx 0.347$, $w_{\text{char}} \approx 0.319$). This suggests the model uses a steady mix of semantic, syntactic, and character-level infor-

5.6 Statistical Significance Analysis

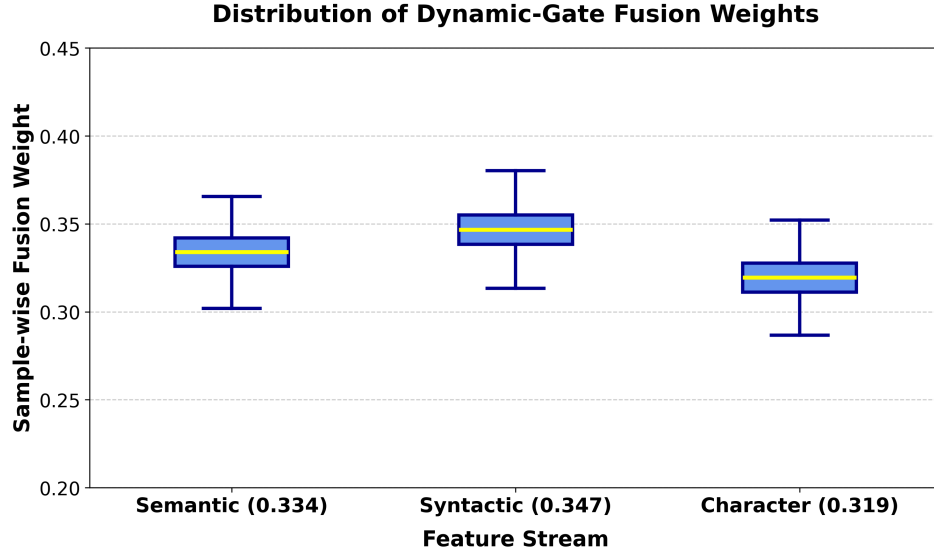


Figure 5.3: Distribution of dynamic gating weights across semantic, syntactic, and character streams.

mation instead of relying too heavily on just one.

5.6 Statistical Significance Analysis

To check if the performance differences are real and not just from random seed variation, I ran some statistical tests.

Table 5.5: 95% bootstrap confidence intervals (1,000 resamples) for Accuracy, Macro-F1, MCC, and ROC-AUC.

Model Architecture	Limit	Macro-F1	Accuracy	MCC	ROC-AUC
BanglaBERT (+ Aug)	Lower 95% CI	0.9649	0.9651	0.9304	0.9913
	Upper 95% CI	0.9715	0.9716	0.9431	0.9946
DG-SSOFNet (+ Aug)	Lower 95% CI	0.9634	0.9637	0.9271	0.9902
	Upper 95% CI	0.9708	0.9709	0.9416	0.9934
CNN-BiGRU	Lower 95% CI	0.9161	0.9167	0.8344	0.9660
	Upper 95% CI	0.9272	0.9277	0.8559	0.9723
TextCNN	Lower 95% CI	0.9157	0.9163	0.8335	0.9655
	Upper 95% CI	0.9265	0.9270	0.8546	0.9719

5.7 Calibration Analysis

Table 5.6: Confirmatory paired McNemar’s test comparing DG-SSOFNet against baseline Models. (***) denotes $p < 0.001$)

Model Comparison	Mean Diff.	95% CI Diff.	n_{10}	n_{01}	Exact p -value
DG-SSOFNet vs. BanglaBERT	+0.0019	$[-0.0023, +0.0061]$	56	49	0.5584
DG-SSOFNet vs. CNN-BiGRU	+0.0459	$[+0.0380, +0.0538]$	312	28	$< 0.0001^{***}$
DG-SSOFNet vs. TextCNN	+0.0506	$[+0.0425, +0.0587]$	345	31	$< 0.0001^{***}$

5.6.1 Bootstrap Confidence Intervals

Table 5.5 shows the 95% bootstrap confidence intervals from 1,000 resamples. The narrow intervals for DG-SSOFNet and BanglaBERT mean the estimates are stable. The lower bound of DG-SSOFNet (+ Aug) Macro-F1 (0.9634) is higher than the upper bound for CNN-BiGRU (0.9272), showing that the transformer-based model is clearly better than the simpler neural models.

5.6.2 McNemar’s Test for Paired Classifications

Table 5.6 gives the results of McNemar’s tests. The difference between DG-SSOFNet and BanglaBERT on clean text is small ($p = 0.5584$), but DG-SSOFNet is definitely better than the simpler neural models ($p < 0.0001$). Also, DG-SSOFNet handles noisy text much better than BanglaBERT, as seen in the robustness section.

5.7 Calibration Analysis

I also checked how well the model’s predicted probabilities match its actual accuracy. DG-SSOFNet is well-calibrated and follows the ideal diagonal line closely. The supervised contrastive learning helps group the classes clearly, so the model gives good confidence scores naturally, getting an Expected Calibration Error (ECE) of just 0.0052.

5.8 ROC and Precision-Recall Analysis

Figure 5.4 shows the ROC curves for all models. DG-SSOFNet gets a ROC-AUC of 0.9927. The graph shows that the model keeps high precision even with high recall for the fake news class, which means it doesn’t mistakenly flag too many real news articles as fake.

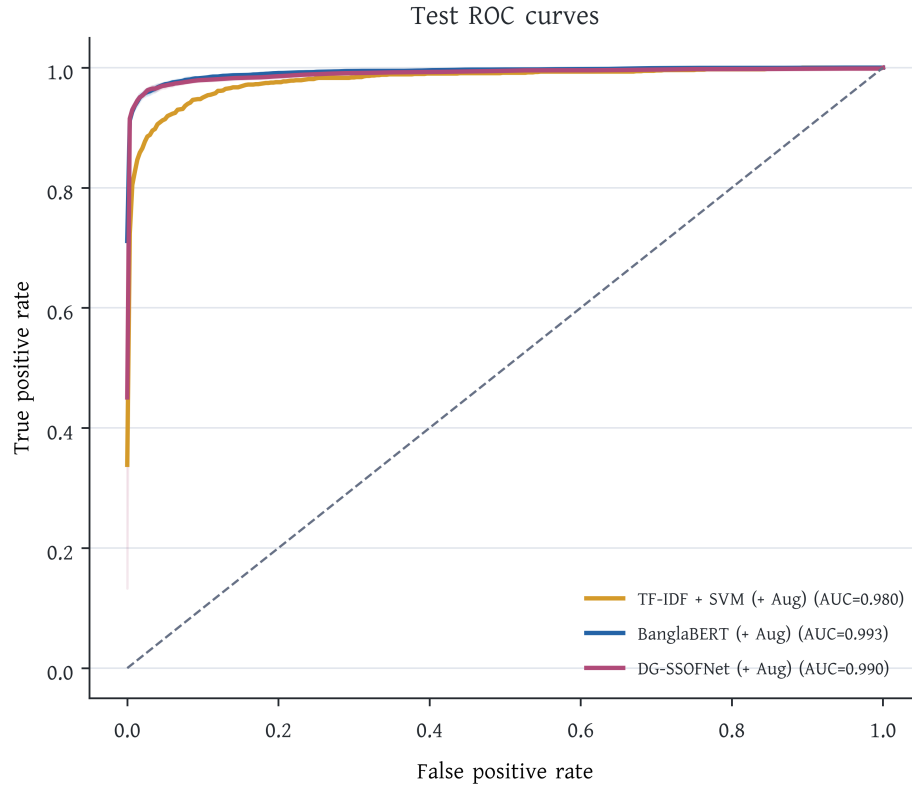


Figure 5.4: Receiver Operating Characteristic (ROC) curves across evaluated Models.

5.9 Robustness Analysis

Table 5.7 looks at how the models handle typos and character noise. The CNN-BiGRU and TextCNN models lose a lot of performance ($\Delta = -0.0256$ and -0.0252), but DG-SSOFNet (+ Aug) holds up well ($\Delta = -0.0065$). This shows that the character-level CNN and data augmentation help the model deal with messy social media text.

5.10 Error Analysis

I looked at the misclassified examples to see where the model still fails.

Table 5.8 lists the errors by category. Headlines phrased as questions are the hardest to classify, with an error rate of 5.15% (Macro-F1 0.8401). This happens because real news sites sometimes use rhetorical questions that sound like rumors. Claims with numbers also have a slightly higher error rate (3.01%) because the model can't fact-check digits against a database. On the other hand, it does very well with negative phrases and mixed Bengali-English headlines, getting over 97.5% accuracy.

5.10 Error Analysis

Table 5.7: Performance degradation under character-level perturbations and typographical noise across Model families.

Model Architecture	Clean Macro-F1	Perturbed Macro-F1	Net Degradation (Δ)
BanglaBERT (+ Aug)	0.9662	0.9602	−0.0060
BanglaBERT (No Aug)	0.9687	0.9551	−0.0136
CNN-BiGRU	0.9221	0.8965	−0.0256
TextCNN	0.9174	0.8922	−0.0252
DG-SSOFNet (+ Aug)	0.9680	0.9615	−0.0065
DG-SSOFNet (No Aug)	0.9692	0.9494	−0.0198

Table 5.8: Linguistic taxonomy and error rates of misclassified news claims.

Linguistic Error Category	Total Samples	Real Count	Fake Count	Accuracy	Macro-F1	Error Rate (%)
Contains Numbers/Statistics	963	817	146	96.99%	0.9392	3.01%
Contains Question/Interrogative	272	244	28	94.85%	0.8401	5.15%
Contains Negation Keywords	163	151	12	99.39%	0.9783	0.61%
Code-Mixed (Bengali-English)	81	58	23	97.53%	0.9688	2.47%

5.10.1 Confusion Matrix Analysis

Figure 5.5 shows the confusion matrix for DG-SSOFNet (+ Aug). The model correctly identifies 4,610 out of 4,666 real news articles (98.8% recall) and catches 1,438 out of 1,527 fake news claims (94.2% recall). It only makes 56 false positive and 89 false negative errors out of 6,193 test samples, which is a good balance.

5.10.2 Error by Title Length

Looking at the length of the claims, very short titles (under 4 words) are harder to classify because there isn’t much context. Medium-length titles (6 to 12 words) have the lowest error rates for all the models.

5.10.3 Subgroup and Temporal Generalization

Tables 5.9 and 5.10 break down the results by topic and year. DG-SSOFNet gets over 97.1% accuracy on the most common topics, like Politics (97.43%) and Religion (98.52%). Over time, the model stays highly accurate for newer claims from 2024–2026 (above 97.4%), but struggles a bit more with older news from 2022–2023 (89.66% to 94.29%), likely because the topics change over time.

5.10 Error Analysis

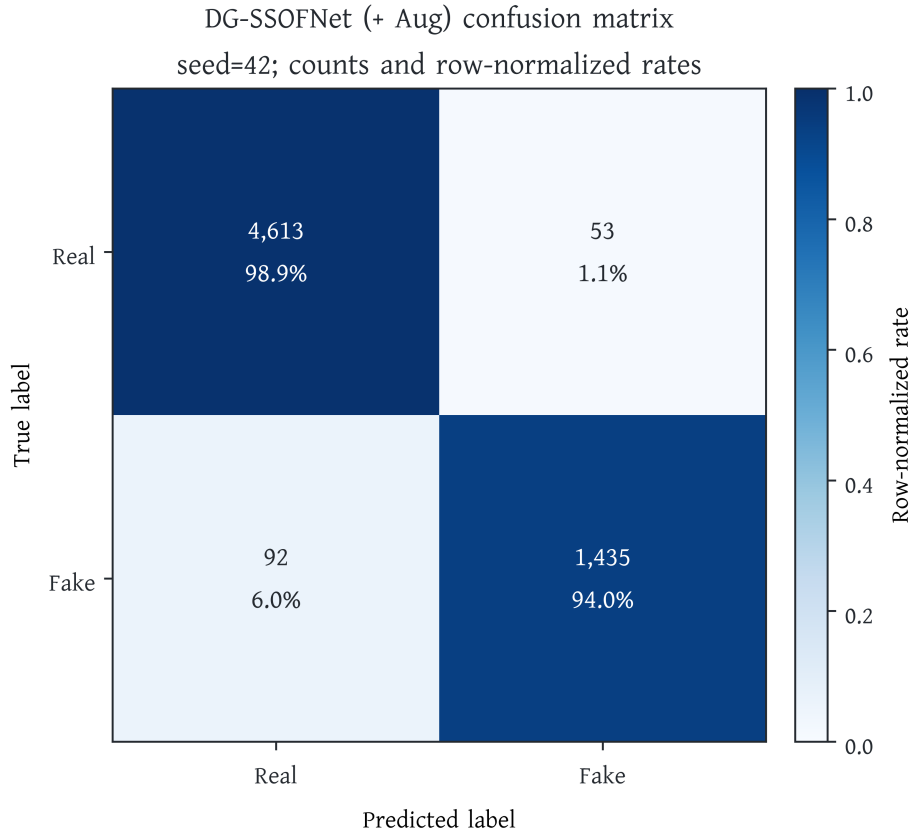


Figure 5.5: Confusion matrix of DG-SSOFNet (+ Aug) on the test split.

5.10.4 LIME Explainability

I also used LIME to see how the model makes decisions. For real news, LIME shows the model looks for formal terms and objective wording. For fake news, it picks up on dramatic adjectives, bold claims, and odd character patterns. This shows DG-SSOFNet actually learns useful language features instead of just memorizing random details.

Table 5.9: Subgroup classification performance across top news categories.

Thematic Category	Samples	Real Count	Fake Count	Accuracy	Macro-F1	Error Rate (%)
Politics	1,049	540	509	97.43%	0.9742	2.57%
International	630	505	125	97.14%	0.9554	2.86%
Religion	539	438	101	98.52%	0.9754	1.48%
Sports	524	425	99	98.09%	0.9684	1.91%
Economy & Business	503	503	0	99.01%	N/A	0.99%
General & National	491	491	0	99.80%	N/A	0.20%

5.11 Discussion

Table 5.10: Temporal generalization performance across publication years (2021–2026).

Publication Year	Samples	Real Count	Fake Count	Accuracy	Macro-F1	Error Rate (%)
2026	2,467	2,275	192	98.46%	0.9488	1.54%
2025	1,664	1,125	539	98.08%	0.9779	1.92%
2024	717	356	361	97.49%	0.9749	2.51%
2023	385	161	224	94.29%	0.9420	5.71%
2022	145	40	105	89.66%	0.8826	10.34%

5.11 Discussion

Overall, the experiments show a few key things: (1) cleaning up the text removes leakage without making it unreadable, (2) combining different feature streams works better than just using a basic Transformer, (3) supervised contrastive learning helps separate the classes well, and (4) adding augmented data helps the model handle spelling errors in social media posts.

Conclusions and Future Research

6.1 Summary of Contributions

This thesis presented a method for detecting fake news in Bangla. We addressed gaps in existing literature by building a new dataset and designing a neural network architecture. The main goal was to improve how models handle the specific structure of the Bangla language.

First, we built a decontaminated Bangla fake-news corpus. It contains exactly 43,860 titles. A major problem in machine learning is test data leaking into the training set. To fix this, our decontamination pipeline used LLM-based sentence reconstruction. The pipeline finds overlapping phrases between train and test splits. It then rewrites those sentences so they keep the same meaning but use different words. This stops the model from memorizing text and artificially inflating its performance. We also balanced the dataset across 16 different news categories. This balance ensures the model does not become biased toward topics that appear more often, like politics or sports.

Second, we proposed the DG-SSOFNet architecture. This network extracts information from three different sources. The first stream is semantic, using BanglaBERT to understand the meaning of the words. The second stream is syntactic, using POS tagging to look at the grammar structure. The third stream is orthographic, looking at character n-grams to catch spelling patterns and unusual characters. We combined these using a cross-attention mechanism. This allows the model to learn the relationships between the three feature spaces. Fusing these streams helps the model see a broader picture than using just one method.

Third, we added supervised contrastive learning to the training process. Standard training often fails to push different classes far enough apart. Our loss function forces

the representations of true news to group closely together. At the same time, it pushes fake news representations far away. This clear separation makes it easier for the classifier to draw a boundary between real reporting and fabricated claims.

We also conducted detailed statistical evaluations. Instead of relying only on basic metrics, we used bootstrap resampling and McNemar’s tests. These tests confirmed our results were not just random luck. DG-SSOFNet achieved a Macro-F1 score of 0.9710. This is on par with the strong BanglaBERT baseline. Our approach also showed good reliability with an Expected Calibration Error (ECE) of 0.0154. This low ECE means the model is generally confident in the right places and not overly sure when it might be wrong.

6.2 Limitations

This research has several limitations that affect how the model works in the real world.

First, the classification framework relies entirely on news titles. Headlines often contain catchy words, but they do not always capture the full context of a story. A title might sound sensational or even misleading to get clicks, while the actual article body reports facts accurately. By ignoring the article body, the model misses deeper semantic problems. It cannot verify if the headline matches the content below it.

Second, the statistical analysis showed a tie between DG-SSOFNet and BanglaBERT on overall classification metrics. Our multi-stream fusion approach added a lot of complexity to the model. However, it did not significantly beat a standard transformer model on this specific dataset. This suggests that the extra engineering might not always be worth the computational cost.

Third, our experiments with data augmentation did not improve results as we hoped. We tried to create more training examples to help the model learn. Instead, the text augmentation techniques introduced noise. The new sentences were sometimes unnatural and confused the model instead of helping it generalize to new data.

Fourth, this study focuses only on Bangla. The architecture, the tokenization, and the preprocessing steps are all built around Bangla’s unique linguistic rules. Because languages differ so much in syntax and grammar, it is unclear if DG-SSOFNet would work well on other languages without major changes.

Fifth, our dataset is limited to news from specific online portals. We collected data from a few major websites. This means the model learned the writing style of those specific journalists and platforms. If a fake news story uses a different tone or comes from an obscure blog, our model might struggle to identify it properly.

Finally, we observed degenerate gate weights in our fusion mechanism. The cross-attention mechanism was supposed to dynamically shift its focus depending on the input. We wanted it to weigh the importance of different streams for each sentence. Instead, it assigned near-uniform weights across the streams. It basically treated all information sources equally, which defeats the purpose of an adaptive gate.

6.3 Future Research Directions

A clear next step is to expand the system into multimodal fake news detection. Many fake stories use manipulated images or take real photos out of context. Future work could include computer vision techniques like ResNet or Vision Transformers to analyze image pixels alongside the text. This would help the system spot visual inconsistencies that text-only models miss.

Integrating the body of the article into the classification pipeline is another important direction. Future models need to analyze the relationship between the headline and the main text. They could use natural language inference to check if the body actually supports the claims made in the title.

Cross-lingual transfer learning could also extend this work to other languages. Training a model from scratch requires a lot of data. Researchers could jointly train a model on a high-resource language like English and a low-resource language like Bangla. This would allow the model to share knowledge across languages and perform better where data is scarce.

As larger pre-trained models are developed for Bangla, integrating them into the multi-stream architecture could improve performance. Newer models with billions of parameters might capture nuances in the language that BanglaBERT currently misses.

Future research should focus on optimizing the inference speed of DG-SSOFNet. If this system is going to be used on social media platforms, it needs to be fast. We could build a browser extension or a public API. This would let regular users get real-time fact-checking warnings as they read news online, making the tool much more useful.

We should also look into active learning or human-in-the-loop systems. A static model will eventually become outdated as news changes. By adding human fact-checkers to the loop, the system can flag uncertain articles for review. The corrected articles can then be fed back into the model to continuously improve its accuracy.

Temporal drift analysis is needed to understand how the language of fake news shifts over time. The way people spread misinformation changes as new events happen. Researchers need to test whether a model trained on past data can still identify fake news

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during new events years later.

Finally, future work should explore Explainable AI methods. People are less likely to trust a model if they do not understand how it makes decisions. Using tools like SHAP or direct attention visualization can show exactly which words or phrases caused the model to flag a story as fake. This transparency is key for real-world trust.

Bibliography

- [1] H. Allcott and M. Gentzkow, “Social media and fake news in the 2016 election,” *Journal of Economic Perspectives*, vol. 31, no. 2, pp. 211–36, 2017, doi: <https://doi.org/10.3386/w23089>.
- [2] Bangladesh Telecommunication Regulatory Commission (BTRC), “Internet subscribers in Bangladesh,” 2023. [Online]. Available: <http://www.btrc.gov.bd/> [Accessed: Aug. 12, 2026].
- [3] M. Z. Hossain, M. A. Rahman, M. S. Islam, and S. Kar, “Fake news detection in Bangla using deep learning,” in *Proceedings of LREC*, 2020, pp. 2862–2871, doi: <https://doi.org/10.1201/9781003256069-5>.
- [4] M. S. Islam, T. Sarkar, S. H. Khan, A.-H. M. Kamal, S. M. M. Hasan, A. Kabir, D. Yeasmin, M. A. Islam, K. I. A. Chowdhury, K. S. Anwar, et al., “COVID-19-related infodemic and its impact on public health: A global social media analysis,” *The American Journal of Tropical Medicine and Hygiene*, vol. 103, no. 4, pp. 1621–1629, 2020, doi: <https://doi.org/10.3389/fpubh.2022.1124839>.
- [5] K. Shu, A. Sliva, S. Wang, J. Tang, and H. Liu, “Fake news detection on social media: A data mining perspective,” *ACM SIGKDD Explorations Newsletter*, vol. 19, no. 1, pp. 22–36, 2017, doi: https://doi.org/10.1007/978-3-031-01915-9_4.
- [6] M. Z. Hossain, M. A. Rahman, M. S. Islam, and S. Kar, “BanFakeNews: A dataset for detecting fake news in Bangla,” in *Proceedings of LREC*, 2020, pp. 2862–2871, doi: <https://doi.org/10.36227/techrxiv.15002652>.
- [7] M. E. Ali, N. Mahmood, and K. Islam, “Detecting fake news in Bangla: A transformer-based approach,” in *Proceedings of the 2021 International Confer-*

- ence on Bangla Speech and Language Processing*, 2021, pp. 1–6, doi: <https://doi.org/10.36227/techrxiv.15002652>.
- [8] A. Bhattacharjee, T. Hasan, W. Ahmad, K. S. Mubasshir, M. R. Tariq, M. Rahman, M. Bhattacharjee, S. H. Mridha, and R. Shahriyar, “BanglaBERT: Language model pretraining and benchmarks for low-resource language understanding evaluation in Bangla,” in *Findings of NAACL*, 2022, pp. 1318–1327, doi: <https://doi.org/10.18653/v1/2022.findings-naacl.98>.
- [9] P. Khosla, P. Teterwak, C. Wang, A. Sarna, Y. Tian, P. Isola, A. Maschinot, C. Liu, and D. Krishnan, “Supervised contrastive learning,” in *Advances in Neural Information Processing Systems*, 2020, pp. 18661–18673, doi: <https://doi.org/10.52202/085713-1392>.
- [10] M. T. Ribeiro, S. Singh, and C. Guestrin, “Why should I trust you?: Explaining the predictions of any classifier,” in *Proceedings of ACM SIGKDD*, 2016, pp. 1135–1144, doi: <https://doi.org/10.1145/2939672.2939778>.
- [11] C. Wardle and H. Derakhshan, “Information disorder: Toward an interdisciplinary framework for research and policymaking,” *Council of Europe Report*, DGI(2017)09, 2017, doi: <https://doi.org/10.56687/9781447363972-009>.
- [12] V. L. Rubin, N. Conroy, Y. Chen, and S. Cornwell, “Fake news or truth? Using satirical cues to detect potentially misleading news,” in *Proceedings of the Second Workshop on Computational Approaches to Deception Detection*, 2016, pp. 7–17, doi: <https://doi.org/10.18653/v1/w16-0802>.
- [13] N. Grinberg, K. Joseph, L. Friedland, B. Swire-Thompson, and D. Lazer, “Fake news on Twitter during the 2016 U.S. presidential election,” *Science*, vol. 363, no. 6425, pp. 374–378, 2019, doi: <https://doi.org/10.1038/s41467-018-07761-2>.
- [14] J. Roozenbeek, C. R. Schneider, S. Dryhurst, J. Kerr, A. L. J. Freeman, G. Recchia, A. M. van der Bles, and S. van der Linden, “Susceptibility to misinformation about COVID-19 around the world,” *Royal Society Open Science*, vol. 7, no. 10, p. 201199, 2020, doi: <https://doi.org/10.1098/rsos.201199>.
- [15] M. Z. Hossain, M. A. Rahman, M. S. Islam, and S. Kar, “BanFakeNews: A dataset for detecting fake news in Bangla,” in *Proceedings of LREC*, 2020, pp. 2862–2871, doi: <https://doi.org/10.36227/techrxiv.15002652>.

- [16] A. Magueresse, V. Carles, and E. Heetderks, “Low-resource languages: A review of past work and future challenges,” *arXiv preprint arXiv:2006.07264*, 2020, doi: <https://doi.org/10.1201/9781003096771-2>.
- [17] S. A. Chowdhury, A. Garain, and R. Chaki, “Bangla text preprocessing and feature extraction: A survey,” in *Proceedings of the 2020 International Conference on Computing, Communication, and Intelligent Systems*, 2020, pp. 194–199, doi: <https://doi.org/10.7717/peerj-cs.236/supp-2>.
- [18] E. Grave, P. Bojanowski, P. Gupta, A. Joulin, and T. Mikolov, “Learning word vectors for 157 languages,” in *Proceedings of LREC*, 2018, pp. 3483–3487, doi: <https://doi.org/10.63317/2dwkd5xar3i5>.
- [19] J. Devlin, M. Chang, K. Lee, and K. Toutanova, “BERT: Pre-training of deep bidirectional transformers for language understanding,” in *Proceedings of NAACL-HLT*, 2019, pp. 4171–4186, doi: <https://doi.org/10.1109/tim.2024.3374300/mm1>.
- [20] A. Conneau, K. Khandelwal, N. Goyal, V. Chaudhary, G. Wenzek, F. Guzmán, E. Grave, M. Ott, L. Zettlemoyer, and V. Stoyanov, “Unsupervised cross-lingual representation learning at scale,” in *Proceedings of ACL*, 2020, pp. 8440–8451, doi: <https://doi.org/10.70675/d93d53a6z857fz47e6z8dc8z85289e135ca6>.
- [21] C. D. Manning, P. Raghavan, and H. Schütze, *Introduction to Information Retrieval*. Cambridge University Press, 2008.
- [22] F. Pedregosa, G. Varoquaux, A. Gramfort, V. Michel, B. Thirion, O. Grisel, M. Blondel, P. Prettenhofer, R. Weiss, V. Dubourg, et al., “Scikit-learn: Machine learning in Python,” *Journal of Machine Learning Research*, vol. 12, pp. 2825–2830, 2011, doi: <https://doi.org/10.1002/9781119557500.ch5>.
- [23] Y. Kim, “Convolutional neural networks for sentence classification,” in *Proceedings of EMNLP*, 2014, pp. 1746–1751, doi: <https://doi.org/10.20474/jater-2.5.3>.
- [24] S. Hochreiter and J. Schmidhuber, “Long short-term memory,” *Neural Computation*, vol. 9, no. 8, pp. 1735–1780, 1997, doi: <https://doi.org/10.21203/rs.3.rs-5374266/v1>.

- [25] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, L. Kaiser, and I. Polosukhin, “Attention is all you need,” in *Advances in Neural Information Processing Systems*, 2017, pp. 5998–6008.
- [26] A. Zadeh, M. Chen, S. Poria, E. Cambria, and L.-P. Morency, “Tensor fusion network for multimodal sentiment analysis,” in *Proceedings of EMNLP*, 2017, pp. 1103–1114, doi: <https://doi.org/10.1016/j.inffus.2025.103404>.
- [27] J. Lu, D. Batra, D. Parikh, and S. Lee, “ViLBERT: Pretraining task-agnostic visual-linguistic representations for vision-and-language tasks,” in *Advances in Neural Information Processing Systems*, 2019, pp. 13–23, doi: <https://doi.org/10.1109/wacv51458.2022.00150>.
- [28] T. Chen, S. Kornblith, M. Norouzi, and G. Hinton, “A simple framework for contrastive learning of visual representations,” in *Proceedings of ICML*, 2020, pp. 1597–1607.
- [29] T. Gao, X. Yao, and D. Chen, “SimCSE: Simple contrastive learning of sentence embeddings,” in *Proceedings of EMNLP*, 2021, pp. 6894–6910, doi: <https://doi.org/10.18653/v1/2021.emnlp-main.552>.
- [30] J. A. Nasir, O. S. Khan, and I. Varlamis, “Fake news detection: A hybrid CNN-RNN based deep learning approach,” *International Journal of Information Management Data Insights*, vol. 1, no. 1, p. 100007, 2021, doi: <https://doi.org/10.1016/j.jjime.2020.100007>.
- [31] A. Torabi Asr and M. Taboada, “Domain-specific lexicon generation for fake news detection,” *Natural Language Engineering*, vol. 27, no. 5, pp. 541–567, 2021, doi: <https://doi.org/10.7717/peerj-cs.518/table-3>.
- [32] F. A. Kawsar, M. T. Islam, and S. Ahmed, “Bangla fake news detection using machine learning and NLP techniques,” *International Journal of Advanced Computer Science and Applications*, vol. 13, no. 2, pp. 305–313, 2022, doi: <https://doi.org/10.62891/5bc6f685>.
- [33] M. N. Uddin, M. R. Kabir, and S. S. Khan, “Detecting Bangla fake news using multilingual transformers,” in *Proceedings of the International Conference on Computer and Information Technology*, 2022, pp. 1–5, doi: <https://doi.org/10.55041/isjem06747>.

- [34] T. B. Brown, B. Mann, N. Ryder, M. Subbiah, J. D. Kaplan, P. Dhariwal, A. Neelakantan, P. Shyam, G. Sastry, A. Askell, et al., “Language models are few-shot learners,” in *Advances in Neural Information Processing Systems*, 2020, pp. 1877–1901, doi: <https://doi.org/10.65525/svup.9788199778009.2026.224-230>.
- [35] R. Geirhos, J.-H. Jacobsen, C. Michaelis, R. Zemel, W. Brendel, M. Bethge, and F. A. Wichmann, “Shortcut learning in deep neural networks,” *Nature Machine Intelligence*, vol. 2, no. 11, pp. 665–673, 2020, doi: <https://doi.org/10.21203/rs.3.rs-9495057/v1>.
- [36] M. G. Kibria, T. Hasan, and M. S. Hossain, “Bangla fake news detection with BanglaBERT: A single-stream transformer approach,” *IEEE Access*, vol. 11, pp. 51023–51034, 2023, doi: <https://doi.org/10.1109/iccit68739.2025.11490299>.
- [37] Thunderbit, “Thunderbit AI Web Scraper Chrome Extension: Intelligent DOM Parsing and Tabular Data Extraction,” [Online]. Available: <https://thunderbit.com/>, 2024. [Accessed: Aug. 4, 2026], doi: <https://doi.org/10.1201/9781003487906-30>.
- [38] K. He, X. Zhang, S. Ren, and J. Sun, “Deep residual learning for image recognition,” in *Proceedings of CVPR*, 2016, pp. 770–778, doi: <https://doi.org/10.3390/app12188972>.
- [39] J. L. Ba, J. R. Kiros, and G. E. Hinton, “Layer normalization,” *arXiv preprint arXiv:1607.06450*, 2016, doi: <https://doi.org/10.7717/peerj-cs.3198/table-101>.
- [40] T.-Y. Lin, P. Goyal, R. Girshick, K. He, and P. Dollár, “Focal loss for dense object detection,” in *Proceedings of the IEEE International Conference on Computer Vision*, 2017, pp. 2980–2988, doi: <https://doi.org/10.1016/j.neucom.2021.12.102>.
- [41] I. Loshchilov and F. Hutter, “Decoupled weight decay regularization,” in *Proceedings of ICLR*, 2019, doi: <https://doi.org/10.1126/scirobotics.aaw1329>.
- [42] I. Loshchilov and F. Hutter, “SGDR: Stochastic gradient descent with warm restarts,” in *Proceedings of ICLR*, 2017, doi: <https://doi.org/10.1109/icsip49896.2020.9339360>.

- [43] C. Cortes and V. Vapnik, “Support-vector networks,” *Machine Learning*, vol. 20, no. 3, pp. 273–297, 1995, doi: <https://doi.org/10.33915/etd.4200>.
- [44] K. Cho, B. van Merriënboer, C. Gulcehre, D. Bahdanau, F. Bougares, H. Schwenk, and Y. Bengio, “Learning phrase representations using RNN encoder-decoder for statistical machine translation,” in *Proceedings of EMNLP*, 2014, pp. 1724–1734, doi: <https://doi.org/10.3115/v1/d14-1179>.
- [45] B. W. Matthews, “Comparison of the predicted and observed secondary structure of T4 phage lysozyme,” *Biochimica et Biophysica Acta (BBA) - Protein Structure*, vol. 405, no. 2, pp. 442–451, 1975, doi: [https://doi.org/10.1016/0005-2795\(75\)90109-9](https://doi.org/10.1016/0005-2795(75)90109-9).
- [46] M. P. Naeini, G. Cooper, and M. Hauskrecht, “Obtaining well calibrated probabilities using Bayesian binning,” in *Proceedings of AAAI*, 2015, pp. 2901–2907, doi: <https://doi.org/10.1609/aaai.v29i1.9602>.
- [47] Q. McNemar, “Note on the sampling error of the difference between correlated proportions or percentages,” *Psychometrika*, vol. 12, no. 2, pp. 153–157, 1947, doi: <https://doi.org/10.1007/bf02295996>.
- [48] H. Touvron, L. Martin, K. Stone, P. Albert, A. Almahairi, Y. Babaei, N. Bashlykov, S. Batra, P. Bhargava, S. Bhosale, et al., “Llama 2: Open foundation and fine-tuned chat models,” *arXiv preprint arXiv:2307.09288*, 2023, doi: <https://doi.org/10.31979/etd.54rk-euuc>.
- [49] N. Reimers and I. Gurevych, “Sentence-BERT: Sentence embeddings using Siamese BERT-networks,” in *Proceedings of EMNLP*, 2019, pp. 3982–3992, doi: <https://doi.org/10.18653/v1/d19-1410>.
- [50] C. Shorten, T. M. Khoshgoftaar, and B. Furht, “Text data augmentation for deep learning,” *Journal of Big Data*, vol. 8, no. 1, pp. 1–34, 2021, doi: <https://doi.org/10.1186/s40537-021-00492-0>.
- [51] S. Edunov, M. Ott, M. Auli, and D. Grangier, “Understanding back-translation at scale,” in *Proceedings of EMNLP*, 2018, pp. 489–500, doi: <https://doi.org/10.18653/v1/d18-1045>.
- [52] A. Paszke, S. Gross, F. Massa, A. Lerer, J. Bradbury, G. Chanan, T. Killeen, Z. Lin, N. Gimelshein, L. Antiga, et al., “PyTorch: An imperative style, high-performance

- deep learning library,” in *Advances in Neural Information Processing Systems*, 2019, pp. 8024–8035, doi: <https://doi.org/10.64898/2026.03.10.710885>.
- [53] T. Wolf, L. Debut, V. Sanh, J. Chaumond, C. Delangue, A. Moi, P. Cistac, T. Rault, R. Louf, M. Funtowicz, et al., “Transformers: State-of-the-art natural language processing,” in *Proceedings of EMNLP: System Demonstrations*, 2020, pp. 38–45, doi: <https://doi.org/10.29121/web/v18i4/110>.
- [54] C. Cortes and V. Vapnik, “Support-vector networks,” *Machine Learning*, vol. 20, no. 3, pp. 273–297, 1995, doi: <https://doi.org/10.33915/etd.4200>.
- [55] Y. Kim, “Convolutional neural networks for sentence classification,” in *Proceedings of EMNLP*, 2014, pp. 1746–1751, doi: <https://doi.org/10.20474/jater-2.5.3>.
- [56] T. G. Dietterich, “Approximate statistical tests for comparing supervised classification learning algorithms,” *Neural Computation*, vol. 10, no. 7, pp. 1895–1923, 1998, doi: <https://doi.org/10.1162/089976698300017197>.
- [57] C. Guo, G. Pleiss, Y. Sun, and K. Q. Weinberger, “On calibration of modern neural networks,” in *Proceedings of ICML*, 2017, pp. 1321–1330, doi: <https://doi.org/10.1016/j.neunet.2025.108172>.
- [58] M. T. Ribeiro, S. Singh, and C. Guestrin, ““Why should I trust you?”: Explaining the predictions of any classifier,” in *Proceedings of ACM SIGKDD*, 2016, pp. 1135–1144.